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Attosecond optical spectroscopy of disordered media

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ABSTRACT

In this thesis, we investigate how structural disorder affects ultrafast electronic dynamics probed in attosecond pump–probe experiments. Using real-time time-dependent density functional theory (RT-TDDFT) simulations performed with the SALMON code, we analyze disorder-induced modifications of carrier photoinjection, intraband current relaxation, and coherent current oscillations in semiconducting silicon.

We demonstrate that disorder reshapes the density of states, reduces the band gap, and introduces elastic scattering that governs momentum relaxation. In an orthogonal pump–probe configuration, the probe-induced intraband current is described within a generalized Drude framework, enabling extraction of the transport relaxation time. The analysis reveals that the effective mass is not constant but depends on the crystal momentum, reflecting the underlying band curvature and its disorder-induced modification.

In the field-free regime, coherent current oscillations are analyzed using an autocorrelation-based approach, allowing a decomposition of dephasing into homogeneous (scattering-induced) and inhomogeneous (dispersion-induced) contributions. A central result is the quantitative link between the homogeneous dephasing time and the transport relaxation time, indicating that the same elastic scattering mechanism controls both intraband transport and interband coherence decay.

Contents

Chapter 1

Introduction

Attosecond optical spectroscopy has opened an unprecedented window into the real-time motion of electrons in matter, enabling direct observation of electronic excitations, charge migration, and correlation-driven dynamics on their natural timescale (10–1000 attoseconds). Over the past decade, techniques such as attosecond transient absorption spectroscopy (ATAS), high-harmonic spectroscopy (HHS), and time-resolved photoemission have revealed rich sub-femtosecond phenomena in atoms, molecules, and crystalline solids. However, despite this rapid progress, the attosecond response of *disordered media* — including liquids, amorphous solids, polymers, and structurally complex hybrid materials — remains far less explored.

Disordered systems are ubiquitous in both natural and technological contexts. Their electronic structure is shaped not only by average bonding environments, but also by stochastic fluctuations of local potentials, short-range structural disorder and localization effects. These features strongly influence ultrafast electron dynamics: they modify mean free paths, induce rapid dephasing, and lead to spatially heterogeneous excitation pathways. Because many relevant photophysical and photochemical processes — photoinduced charge transfer, interfacial electron injection, initial steps of bond rearrangements — occur in such irregular environments, understanding attosecond-scale dynamics in disordered media is essential for accurate interpretation of measurements and rational materials design.

Recent experimental studies have demonstrated that attosecond techniques are indeed sensitive to disorder-induced scattering. High-harmonic spectroscopy in liquids has been shown to encode electron mean free paths and dephasing rates in the shape and phase of the emitted harmonics [1]. Attosecond measurements in solids have captured ultrafast modifications of the band structure during strong-field excitation [2]. These results motivate the development of quantitative, predictive theoretical frameworks capable of describing attosecond processes in non-crystalline environments.

As will be shown below, incorporating structural disorder in simulations naturally gives rise to ultrafast electronic dephasing, a process that plays a central role in attosecond experiments. Independent experimental evidence for such rapid relaxation already exists in ultrafast spectroscopy of crystalline materials. Phase-resolved transient absorption measurements in silicon driven by sub-5-fs excitation pulses reveal extremely rapid carrier dynamics, including single-electron momentum relaxation times of approximately 10 fs and Drude collision times that reach only a few femtoseconds at the earliest delays. These experiments further report a transiently evolving optical effective mass during the initial stages of carrier relaxation, highlighting the strongly time-dependent character of electronic transport immediately following photoexcitation [3].

Direct information about electronic coherence can also be obtained from high-harmonic spectroscopy. High-harmonic polarimetry measurements in strongly driven two-dimensional materials demonstrate that the polarization state of the emitted harmonics encodes the underlying electron–hole coherence. From these measurements, an electron–hole dephasing time T_2 of only a few femtoseconds was extracted [4]. Although performed in a different material platform, these results provide compelling evidence that HHG observables can be governed by femtosecond-scale coherence loss.

The necessity of such short coherence times is also well established in theoretical descriptions of solid-state high-harmonic generation. Reciprocal-space simulations typically require phenomenological dephasing times on the order of a few femtoseconds in order to reproduce experimentally observed harmonic spectra. Recent work has clarified the physical origin of these apparently extreme parameters by analyzing HHG dynamics in real space: strong dephasing suppresses recombination events involving large electron–hole separations and thereby eliminates unrealistically large contributions from long electron trajectories [5].

Complementary modeling strategies introduce explicit decoherence mechanisms into strong-field simulations, for example through phenomenological imaginary potentials that mimic scattering-induced coherence loss. Such approaches successfully reproduce key experimental trends in solid-state HHG and further demonstrate that femtosecond-scale decoherence constitutes a critical control parameter governing the nonlinear optical response of strongly driven solids [6].

Taken together, these experimental and theoretical studies indicate that ultrafast dephasing and scattering are intrinsic features of strongly driven electronic systems rather than merely phenomenological modeling parameters. In structurally disordered media, where fluctuations of the local potential introduce additional scattering channels and spatially heterogeneous electronic environments, coherence loss is expected to become even more pronounced. Accurately describing these processes therefore represents a central challenge for theoretical treatments of attosecond spectroscopy in non-crystalline materials.

From a theoretical perspective, modeling attosecond optical responses of disordered systems requires a method that simultaneously accounts for (i) nonperturbative, real-time electronic dynamics and (ii) the explicit atomic-scale complexity of the medium. Real-time time-dependent density functional theory (RT-TDDFT) satisfies these requirements and has become a central tool for first-principles simulations of strong-field and attosecond phenomena in both molecules and solids. RT-TDDFT allows direct propagation of the electronic state in time, naturally capturing nonlinear responses, transient absorption signals, and high-harmonic emission. However, applying RT-TDDFT to disordered media remains challenging because of the need for large supercells, ensemble averaging over structural configurations, and the treatment of ultrafast scattering and decoherence — effects that are either absent or strongly suppressed in periodic crystals.

The objective of this thesis is therefore to develop, validate, and apply a real-time TDDFT-based methodology for attosecond optical spectroscopy of disordered media. This includes constructing representative structural ensembles, implementing appropriate averaging procedures, computing ATAS and HHS observables, and identifying spectral signatures of disorder such as enhanced scattering, ultrafast dephasing and relaxation. By comparing theoretical predictions with available experimental results, this work aims to clarify the microscopic origin of attosecond signals in non-crystalline environments and to provide practical guidelines for future attosecond spectroscopy of complex materials.

1.1 Thesis Structure

This thesis is organized to provide a systematic progression from theoretical foundations to microscopic modeling and, ultimately, to the analysis of ultrafast observables in disordered silicon.

Chapter 2 introduces the real-time time-dependent density functional theory (RT-TDDFT) framework that underlies all simulations presented in this work. The time-dependent Kohn–Sham formalism, exchange–correlation approximations, and the implementation within the SALMON code are described in detail. The temporal validity window of RT-TDDFT for attosecond pump–probe spectroscopy in solids is also discussed, clarifying which physical processes are captured within a mean-field description and which require methods beyond adiabatic TDDFT.

Chapter 3 is devoted to structural disorder. Different models of displacement and amorphous disorder are introduced, together with their implementation in supercell simulations. The relationship between structural fluctuations, modification of the electronic potential landscape, and elastic scattering processes is established, forming the basis for the later analysis of transport and coherence decay.

Chapter 4 focuses on silicon as the material platform for this study. Its crystal structure and band structure are reviewed, with particular emphasis on symmetry properties near the Γ point. The second part of this chapter presents a detailed analysis of the electronic density of states (DOS) under increasing disorder, highlighting band-gap renormalization, smoothing of van Hove singularities, and the emergence of exponential band-edge tails.

Having established the structural and electronic groundwork, Chapter 5 investigates ultrafast carrier photoinjection induced by few-femtosecond pump pulses. By systematically varying photon energy and disorder strength, different excitation regimes — from strongly nonlinear multiphoton processes to single-photon interband transitions — are identified and connected to the evolving band structure.

Chapter 6 analyzes current dynamics during the probe pulse in an orthogonal pump–probe configuration. This setup enables a clear separation between carrier generation and transport. The probe-induced intraband current is described within a generalized Drude framework, allowing extraction of the momentum relaxation time and examination of effective-mass variations in reciprocal space.

Chapter 7 examines coherent current oscillations (CCO) in the field-free regime following the pump pulse. Using a semiconductor Bloch equation perspective combined with autocorrelation-based coherence analysis, homogeneous (scattering-induced) and inhomogeneous (dispersion-induced) dephasing mechanisms are disentangled. A quantitative link between transport relaxation time and homogeneous coherence decay is established.

Finally, Chapter 8 integrates the results of all previous chapters. A unified physical picture is developed in which density-of-states modification, photoinjection efficiency, transport relaxation, and coherence decay are shown to originate from the same disorder-induced potential fluctuations. The broader implications for attosecond spectroscopy and ultrafast solid-state physics are discussed.

Chapter 2

Real-Time Time-Dependent Density Functional Theory

2.1 Theory and Principles of RT-TDDFT

Real-time time-dependent density functional theory (RT-TDDFT) provides an *ab initio* framework for simulating the response of interacting electrons to time-dependent electromagnetic fields. In contrast to frequency-domain (Casida-type) TDDFT, where excitation energies are obtained from an eigenvalue problem formulated in the frequency domain, RT-TDDFT directly propagates the Kohn–Sham (KS) orbitals in real physical time. This formulation makes it particularly well suited for describing ultrafast and nonlinear phenomena, including strong-field dynamics and attosecond pump–probe processes in solids [7, 8, 9].

2.1.1 Time-Dependent Kohn–Sham Equations

The theoretical foundation of RT-TDDFT is the Runge–Gross theorem [7], which establishes a one-to-one correspondence between the time-dependent electron density $n(\mathbf{r}, t)$ and the external potential, up to a purely time-dependent gauge function, for a fixed initial many-body state. This allows the interacting system to be mapped onto an auxiliary non-interacting KS system whose orbitals satisfy

$$i\frac{\partial}{\partial t}\psi_{n\mathbf{k}}(\mathbf{r}, t) = \hat{H}_{\mathrm{KS}}[n](t)\,\psi_{n\mathbf{k}}(\mathbf{r}, t), \tag{2.1}$$

with the Hamiltonian

$$\hat{H}_{\mathrm{KS}}(t) = \frac{1}{2}\left[-i\nabla + \mathbf{A}(t)\right]^2 + v_{\mathrm{ion}}(\mathbf{r}) + v_{\mathrm{H}}[n](\mathbf{r}, t) + v_{\mathrm{xc}}[n](\mathbf{r}, t). \tag{2.2}$$

Here v_{ion} denotes the ionic potential, typically represented by norm-conserving pseudopotentials, v_{H} is the Hartree potential, and v_{xc} is the exchange–correlation (XC) potential. The external laser field is introduced via the time-dependent vector potential $\mathbf{A}(t)$ in the velocity gauge, which is equivalent to a scalar potential formulation in the length gauge [8, 10].

The time-dependent electron density is reconstructed from the occupied KS orbitals as

$$n(\mathbf{r}, t) = \sum_{n\mathbf{k}} f_{n\mathbf{k}}\left|\psi_{n\mathbf{k}}(\mathbf{r}, t)\right|^2, \tag{2.3}$$

where $f_{n\mathbf{k}}$ are the occupation numbers of the initial ground state. In practical implementations, Eqs. (2.1)–(2.3) are discretized either on a real-space grid or in a plane-wave basis and propagated using unitary time-evolution schemes such as enforced time-reversal symmetry (ETRS) or midpoint propagators [8, 10].

2.1.2 Exchange–Correlation Potentials

A central approximation in practical RT-TDDFT is the choice of the time-dependent XC potential. In most applications, including the present work, the adiabatic approximation is employed, in which the ground-state XC functional is evaluated at the instantaneous density. This neglects memory effects but has proven reliable for early-time coherent dynamics in solids [11, 9].

Within ground-state DFT, several families of functionals are commonly used. The local density approximation (LDA) expresses the XC energy as a local function of the density and is typically parameterized using homogeneous electron gas data, such as in the Perdew–Zunger (PZ) form. Generalized gradient approximations (GGA) extend this by incorporating density gradients and often improve structural properties. More advanced meta-GGA-type functionals, including the Tran–Blaha modified Becke–Johnson (TB-mBJ) potential, incorporate kinetic-energy density or orbital-dependent information and can significantly improve band-gap predictions in semiconductors.

For extended systems, the choice of functional directly influences the band structure, transition energies, and effective masses, and therefore impacts the simulated ultrafast response.

2.1.3 Exchange–Correlation Potentials in SALMON

All RT-TDDFT simulations in this thesis are performed using the SALMON (Scalable *Ab initio* Light–Matter Simulator for Optics and Nanoscience) code [12]. SALMON is a real-space, real-time TDDFT package designed for molecules, nanostructures, and periodic solids, and is efficiently parallelized for large-scale simulations.

In SALMON, the XC functional is specified through the `&functional` input namelist [13]. For periodic systems such as crystalline silicon, the most relevant adiabatic choices include the Perdew–Zunger LDA (PZ-LDA), which is computationally efficient and used throughout this work; a modified PZ form (PZM-LDA); and the TB-mBJ meta-GGA potential combined with Perdew–Wang correlation, which is known to yield improved band gaps in semiconductors.

2.2 Ultrafast Electron Dynamics in Silicon and the Validity Window of RT-TDDFT

To assess what aspects of silicon's ultrafast response can be reliably described within RT-TDDFT, it is instructive to consider the hierarchy of microscopic processes triggered by a few-femtosecond excitation pulse and to analyze how these processes are represented within the time-dependent KS framework.

Immediately during the pump pulse, coherent interband excitation generates polarization and redistributes population between valence and conduction bands. Attosecond transient absorption experiments in crystalline silicon resolve step-like changes in absorption synchronized with the electric-field half-cycles, with rise times on the order of several hundred attoseconds [2]. RT-TDDFT accurately captures this early-time coherent regime, including sub-femtosecond population transfer and the onset of carrier-induced band-gap renormalization [2, 8, 9].

On time scales of approximately 1–5 fs, injected carriers screen the Coulomb interaction and modify the KS potential, producing transient band-gap shifts and spectral broadening. Within RT-TDDFT, this effect arises from the time-dependent Hartree and XC potentials and reproduces the temporal evolution observed experimentally, although correlation effects beyond the adiabatic approximation may affect quantitative accuracy [2, 11].

During the pulse, intraband acceleration and strong-field effects also occur. These include dynamical Stark and Franz–Keldysh shifts as well as high-harmonic generation in sufficiently strong fields. Such phenomena are naturally described in RT-TDDFT and have been extensively modeled within this framework [8, 14, 9].

In disordered silicon, structural fluctuations introduce distributions of local transition energies and effective band gaps. Even in the absence of explicit scattering, this static inhomogeneity leads to dephasing of the macroscopic polarization on time scales of several femtoseconds. Because RT-TDDFT explicitly incorporates the real-space structure of the supercell, such inhomogeneous broadening is inherently included in simulations of disordered systems.

Beyond these coherent processes, electron–electron scattering and carrier thermalization gradually redistribute energy and momentum among carriers. While instantaneous Coulomb interactions are fully included through the Hartree and XC potentials, true collisional processes associated with imaginary parts of many-body self-energies are not explicitly treated in adiabatic RT-TDDFT. Many-body approaches such as nonequilibrium Green's functions indicate that electron–electron thermalization in silicon occurs on multi-femtosecond time scales extending to several tens of femtoseconds [15, 16, 17, 18]. RT-TDDFT therefore provides an accurate description of the coherent and quasi-collisionless regime, but does not capture irreversible thermalization.

On longer time scales, electron–phonon coupling drives momentum relaxation and carrier cooling, typically within tens to hundreds of femtoseconds [18, 19, 3, 20]. Such processes require explicit lattice dynamics and are beyond the scope of the present simulations.

Taken together, these considerations define a natural validity window for adiabatic RT-TDDFT in silicon. The method is quantitatively reliable for the early-time coherent regime up to approximately 20–30 fs, encompassing interband excitation, mean-field screening, strong-field intraband dynamics, and disorder-induced inhomogeneous dephasing. At longer delays, RT-TDDFT continues to provide qualitatively meaningful insight into the coherent component of the response, but a fully quantitative treatment of irreversible relaxation requires approaches beyond mean-field TDDFT.

Since the present thesis focuses on attosecond pump–probe spectroscopy with pulse durations of a few femtoseconds and delay times within several tens of femtoseconds, the key observables lie predominantly within this coherent validity window. RT-TDDFT as implemented in SALMON is therefore well suited to capture the essential physics addressed in this work.

Chapter 3

Disorder in Solids

3.1 Types of Disorder

Disorder in solids refers to any deviation from perfect periodicity in the atomic arrangement. This can significantly alter electronic, optical, and transport properties, because the breaking of translational symmetry allows new scattering processes, modifies the density of states, and affects carrier dynamics [21, 22].

Disorder can be classified into several types:

- **Substitutional disorder:** some lattice atoms are replaced by different species, modifying the local potential.
- **Interstitial disorder:** extra atoms occupy positions between the lattice sites.
- **Vacancies:** missing atoms that locally disrupt the potential.
- **Random displacements of atoms:** atoms are shifted randomly from their equilibrium lattice positions without changing the overall crystal structure.
- **Amorphous disorder:** no long-range order; the atomic positions are random beyond short-range correlations.

In this work, we focus on two cases in silicon:

1. **Random atomic displacements** within the crystal lattice. This can be treated using standard scattering theory (Born approximation) and leads to finite lifetimes, spectral broadening, and band shifts. 2. **Amorphous silicon**, which is strongly disordered and exhibits Anderson-type localization. In this regime, the absence of translational symmetry prevents a description in terms of weak perturbations of Bloch states and leads to profound modifications of transport properties and optical response [23, 24, 25].

Disorder affects observable properties such as band structure and density of states (DOS), spectral broadening and self-energy corrections, optical absorption and high-harmonic generation, carrier transport and relaxation times.

3.2 Static Disorder and Scattering Theory

In systems with weak random atomic displacements, the electron dynamics can be treated using the widely used effective-mass approximation as a simple theoretical description [26]. The dynamics are governed by the Hamiltonian

$$H = -\frac{\hbar^2}{2m^*}\nabla^2 + \delta V(\mathbf{r}), \tag{3.1}$$

where m^* is the conduction-band effective mass and $\delta V(\mathbf{r})$ represents the disorder potential produced by stochastic atomic displacements.

The displacement of an atom from its equilibrium position modifies the local potential. In the regime of small atomic shifts, the perturbation can be modeled as a short-range scatterer. A widely used approximation in scattering theory [21, 24] is to represent each displacement-induced perturbation by a delta-function potential,

$$\delta V(\mathbf{r}) = \sum_n V_0\, \delta(\mathbf{r} - \mathbf{R}_n), \tag{3.2}$$

where V_0 characterizes the strength of the displacement-induced potential. Such delta-like representations are routinely used in analytical disorder models [27].

The matrix element entering the Born approximation is [21]

$$|\delta V_{\mathbf{k}\mathbf{k}'}|^2 \equiv |\langle \mathbf{k}'|\delta V|\mathbf{k}\rangle|^2 = \frac{V_0^2}{V^2}, \tag{3.3}$$

where the states $|\mathbf{k}\rangle$ and $|\mathbf{k}'\rangle$ are plane waves. So, scattering is isotropic and independent of the momenta.

In a parabolic approximation

$$E_{\mathbf{k}} = E_c + \frac{\hbar^2 k^2}{2m^*}, \tag{3.4}$$

where E_c is the conduction band minimum and the free retarded Green's function [24]

$$G_0(E, \mathbf{k}) = \frac{1}{E - E_{\mathbf{k}} + i0^+}. \tag{3.5}$$

In single-particle scattering theory, the disorder-averaged self-energy in the Born approximation [21, 28] is

$$\Sigma(E) = n_i V_0^2 \int \frac{d^3k}{(2\pi)^3} \frac{1}{E - E_{\mathbf{k}} + i0^+}, \tag{3.6}$$

where n_i is the concentration of the scattering centers. Using the effective-mass DOS [22]

$$\rho_0(E) = \frac{1}{2\pi^2} \left(\frac{2m^*}{\hbar^2} \right)^{3/2} \sqrt{E - E_c}\, \Theta(E - E_c), \tag{3.7}$$

Eq. (3.6) yields [21]

$$\Im\Sigma(E) = -\pi n_i V_0^2 \rho_0(E), \qquad \Re\Sigma(E) = n_i V_0^2\, \mathcal{P} \int dE' \, \frac{\rho_0(E')}{E - E'}. \tag{3.8}$$

The disorder-induced linewidth is therefore

$$\Gamma(E) = -2\Im\Sigma(E) = 2\pi n_i V_0^2 \rho_0(E), \tag{3.9}$$

an expression standard in scattering theory [21].

Because $\rho_0(E) \propto \sqrt{E - E_c}\, \Theta(E - E_c)$, the width increases as

$$\Gamma(E) \propto \sqrt{E - E_c}\, \Theta(E - E_c). \tag{3.10}$$

The disorder-averaged Green's function becomes

$$G(E, \mathbf{k}) = \frac{1}{E - E_{\mathbf{k}} - \Sigma(E)} = \frac{1}{E - E_{\mathbf{k}} - \Re\Sigma(E) + i\Gamma(E)/2}, \tag{3.11}$$

and the spectral function is [21]

$$A(E, \mathbf{k}) = -\frac{1}{\pi} \Im G(E, \mathbf{k}) = \frac{1}{\pi} \frac{\Gamma(E)/2}{\left(E - E_{\mathbf{k}} - \Re\Sigma(E)\right)^2 + (\Gamma(E)/2)^2}. \tag{3.12}$$

The scattering rate follows directly:

$$\frac{1}{\tau(E)} = \frac{\Gamma(E)}{\hbar} = \frac{2\pi}{\hbar} n_i V_0^2 \rho_0(E), \tag{3.13}$$

which is the standard Fermi golden rule for short-range impurities [21, 22].

The transport time is given by [21]

$$\frac{1}{\tau_{\text{tr}}(E)} = \frac{2\pi}{\hbar} n_i V_0^2 \int \frac{d^3k'}{(2\pi)^3}(1-\cos\theta_{\mathbf{kk'}})\delta(E_{\mathbf{k}} - E_{\mathbf{k'}}). \tag{3.14}$$

For isotropic δ-scatterers,

$$\tau_{\text{tr}}(E) = \tau(E), \tag{3.15}$$

as discussed in [21, 27].

The disorder-modified DOS is

$$\rho(E) = \int \frac{d^3k}{(2\pi)^3} A(E, \mathbf{k}), \tag{3.16}$$

resulting in a broadened square-root onset at the band edge [29].

3.2.1 Analytical Density of States for Parabolic Band with Scattering Broadening

The clean DOS is [29]

$$\rho_0(E) = \frac{1}{2\pi^2}\left(\frac{2m^*}{\hbar^2}\right)^{3/2} \sqrt{E - E_c}\,\Theta(E - E_c). \tag{3.17}$$

To account for disorder-induced lifetime effects, we use the standard approximation from many-body scattering theory [21, 24]

$$\Sigma(E) = \Delta - i\Gamma/2$$

considering Δ and Γ as constants, so that effectively

$$E \to E - \Delta + i\Gamma/2.$$

From general considerations, Δ is of the order of Γ with an accuracy of logarithmic trimming. If we make such a substitution in our previous formula for the case without defects we get the broadened DOS as the real part of the clean DOS evaluated at complex energy [21]:

$$\rho(E) = \Re\left[C\sqrt{(E - E_c - \Delta) + i\frac{\Gamma}{2}}\right], \tag{3.18}$$

with

$$C = \frac{1}{2\pi^2}\left(\frac{2m^*}{\hbar^2}\right)^{3/2}.$$

A more rigorous derivation can be found in the A.1.

Here we defined $E_c^* = E_c + \Delta$ Let $u = E - E_c^*$, $v = \Gamma/2$, and $R = \sqrt{u^2 + v^2}$. Using the standard expression for complex square roots,

$$\sqrt{u + iv} = \sqrt{\frac{R+u}{2}} + i\sqrt{\frac{R-u}{2}},$$

the broadened DOS becomes

$$\rho(E) = C\sqrt{\frac{\sqrt{(E - E_c^*)^2 + (\Gamma/2)^2} + (E - E_c^*)}{2}}. \tag{3.19}$$

This expression reduces to the clean DOS (3.17) when $\Gamma \to 0$, and produces a finite sub-gap tail characteristic of disorder broadening. Thus we can say that we have a broadening of the order of Γ. However, the nature of broadening in real silicon samples (especially in amorphous ones) is much more complex and has an exponential character.

3.3 Dataset of Amorphous Silicon Structures

In this work we will use dataset of amorphous structures from [30].

To characterize atomic environments, Rosset *et al.* employ the Smooth Overlap of Atomic Positions (SOAP) descriptor.

For each atom i, we define a neighbor density

$$\rho_i(\mathbf{r}) = \sum_j \exp\left(-\frac{|\mathbf{r}-\mathbf{r}_{ij}|^2}{2\sigma^2}\right) f_c(|\mathbf{r}_{ij}|), \tag{3.20}$$

where the sum runs over neighbors j within the cutoff radius r_c. Here σ is the Gaussian smearing width and $f_c(r)$ is a smooth cutoff function.

Then we expand the density (3.20) in orthonormal radial functions $g_n(r)$ and spherical harmonics $Y_{\ell m}$:

$$\rho_i(\mathbf{r}) = \sum_{n\ell m} c^{(i)}_{n\ell m}\, g_n(r)\, Y_{\ell m}(\hat{\mathbf{r}}), \tag{3.21}$$

where $c^{(i)}_{n\ell m}$ are expansion coefficients. Rotational invariance is achieved by constructing the SOAP power spectrum

$$p^{(i)}_{nn'\ell} = \sum_m c^{(i)*}_{n\ell m} c^{(i)}_{n'\ell m}, \tag{3.22}$$

which serves as the rotationally invariant descriptor vector $\mathbf{p}^{(i)}$.

Similarity between environments i and k is quantified using the SOAP kernel

$$k_{\mathrm{SOAP}}(i,k) = \left(\frac{\mathbf{p}^{(i)}\cdot\mathbf{p}^{(k)}}{\|\mathbf{p}^{(i)}\|\,\|\mathbf{p}^{(k)}\|}\right)^{\zeta}, \tag{3.23}$$

where $\zeta \geq 1$ controls the sharpness of the similarity measure. By construction,

$$0 \leq k_{\mathrm{SOAP}}(i,k) \leq 1,$$

with $k_{\mathrm{SOAP}} = 1$ only for identical local structures (up to rotation). Rosset *et al.* use $k_{\mathrm{SOAP}}(i, \mathrm{diamond})$ to quantify how closely a given local environment resembles crystalline diamond-structure silicon, enabling continuous discrimination between amorphous, paracrystalline, and crystalline domains.

Chapter 4

Silicon

4.1 Silicon as a Semiconductor for Attosecond Spectroscopy

Silicon is the prototypical covalent semiconductor and, in many respects, the "hydrogen atom" of solid-state physics: virtually every new band-structure or many-body method is first benchmarked on crystalline Si [29, 31]. For attosecond optical spectroscopy, silicon provides a conceptually clean yet physically rich platform. It combines a well-characterized covalent band structure with an indirect band gap, multiple equivalent conduction-band valleys, substantial electron–phonon coupling, and a technologically important disordered and amorphous phase.

In the following, we summarize the key features of crystalline and disordered silicon that are most relevant for attosecond pump–probe experiments. The discussion focuses on the formation of valence and conduction bands from atomic orbitals, the special role of the Γ point in the Brillouin zone, and the way static defects and amorphous disorder modify the band structure and thereby influence ultrafast dynamics.

4.1.1 Crystal Structure and Formation of the Band Structure

Crystalline silicon adopts the diamond structure, which can be viewed as a face-centered cubic (fcc) Bravais lattice with a two-atom basis. Each Si atom forms four covalent bonds in a tetrahedral sp^3 configuration. The strong directional character of these sp^3 bonds leads to a bonding manifold forming the valence bands and an antibonding manifold forming the conduction bands, separated by a band gap between predominantly bonding and antibonding states [31]. The valence bands derive mainly from hybridized $3s$ and $3p$ atomic orbitals, while the conduction bands originate from their antibonding counterparts.

The Brillouin zone (BZ) of the fcc lattice contains the high-symmetry points Γ (zone center), X (face centers), L (zone corners), and the Δ lines connecting Γ and X along $\langle 100 \rangle$ directions. In silicon, the valence-band maximum (VBM) is located at Γ, whereas the conduction-band minima (CBM) lie in six equivalent Δ valleys situated approximately 0.85 of the way from Γ to X along each $\langle 100 \rangle$ direction [32, 33]. As a consequence, silicon is an indirect-gap semiconductor: the fundamental band gap connects electronic states with different crystal momenta.

At room temperature, the indirect gap is approximately $1.12\,\mathrm{eV}$, whereas the lowest direct gap at Γ is substantially larger, about $3.4\,\mathrm{eV}$ [31, 32]. Optical transitions near the indirect band edge therefore require phonon assistance in bulk Si, while purely vertical (direct) transitions at Γ occur only at higher photon energies.

4.1.2 Electronic Structure near the Γ Point

Despite silicon's indirect band gap, the Γ point ($\mathbf{k} = 0$) plays a central role in attosecond spectroscopy. The valence-band maximum resides at Γ, optical matrix elements are typically strongest near this point, and the symmetry classification of states is most transparent there [31, 35]. For ultrashort optical excitation, which predominantly induces vertical transitions in $\mathbf{k}$-space, the Γ-point picture provides a natural starting framework.

At Γ, the top of the valence band originates from three degenerate p-like orbitals (p_x, p_y, p_z) combined with electron spin. Spin–orbit coupling couples orbital angular momentum $L = 1$ with

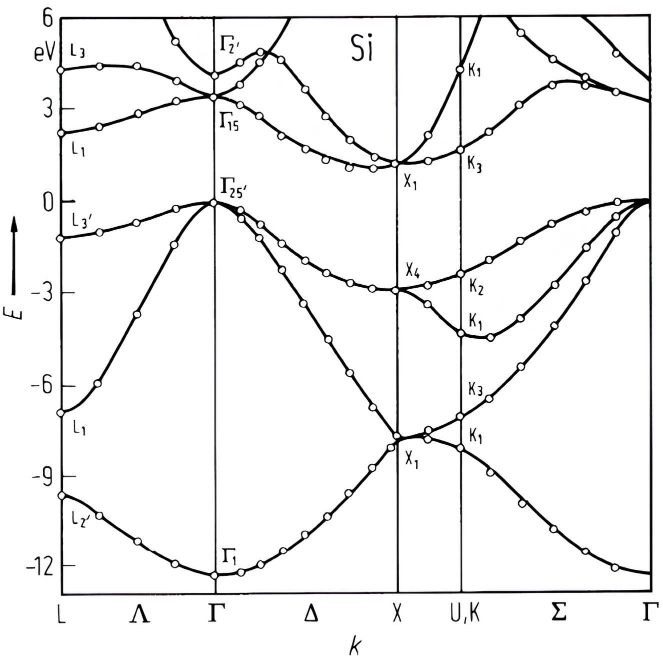

Figure 4.1: Electronic band structure of crystalline silicon, highlighting the indirect fundamental gap between the valence-band maximum at Γ and the conduction-band minima in the Δ valleys. Taken from [34].

spin $S = 1/2$, resulting in total angular momentum manifolds with $J = 3/2$ and $J = 1/2$. The $J = 3/2$ states form a fourfold-degenerate manifold at the valence-band maximum, while the $J = 1/2$ split-off band lies approximately 44 meV below in silicon [31].

Within the $J = 3/2$ manifold, the states are conventionally labeled as heavy-hole (HH) and light-hole (LH) bands, corresponding to different projections of total angular momentum. Although degenerate at Γ, these bands differ in curvature: the HH band is flatter and thus associated with a larger effective mass, while the LH band is more dispersive and lighter. For finite $\mathbf{k}$, these bands split and mix, but for the small-$|\mathbf{k}|$ regions relevant to ultrafast optical excitation, the Γ-point classification remains physically meaningful.

The lowest conduction-band state at Γ is predominantly s-like and defines the bottom of the direct gap. Although this state lies higher in energy than the Δ-valley minima that define the indirect gap, it plays a crucial role in direct optical transitions and in strong-field excitation scenarios [31]. For sufficiently intense few-cycle pulses, multiphoton and strong-field processes can inject carriers into Γ-like conduction states even when the photon energy is below the direct gap.

Band-structure calculations show that the conduction-band dispersion near Γ is relatively isotropic and approximately parabolic at low energies [32, 36]. This simplifies the modeling of early-time carrier dynamics in $\mathbf{k}$-space. In the regime considered in this thesis, where few-femtosecond pump pulses induce predominantly vertical excitation, initial carrier injection can be approximated as transitions from HH/LH valence states at Γ into conduction states at nearly the same crystal momentum. Subsequent redistribution of carriers into the Δ valleys occurs on much longer time scales through electron–phonon scattering [18, 15].

On attosecond to few-femtosecond time scales, the carrier population therefore remains largely localized in the vicinity of the initial injection region in reciprocal space, making a Γ-centered description both convenient and physically justified [2].

4.2 Density of States

The electronic density of states (DOS) provides a fundamental characterization of how structural disorder modifies the electronic structure of silicon. Since ultrafast excitation and transport processes are governed by the available phase space and band dispersion, understanding how disorder reshapes the DOS is essential for interpreting both photoinjection efficiency and subsequent carrier dynamics.

Within RT-TDDFT, the DOS is extracted from the Kohn–Sham eigenvalues obtained for each supercell configuration. While Kohn–Sham eigenvalues do not formally represent quasiparticle energies, they provide a reliable qualitative description of disorder-induced band broadening and gap modification [37].

All calculations here and in subsequent sections were performed using a $3 \times 3 \times 3$ supercell for both amorphous and displacement disorder, which was sufficient to ensure convergence. Each primitive cell contains eight silicon atoms. We took a real-space grid of $54 \times 54 \times 54$ points and a $5 \times 5 \times 5$ k-point mesh which yielded well-converged results.

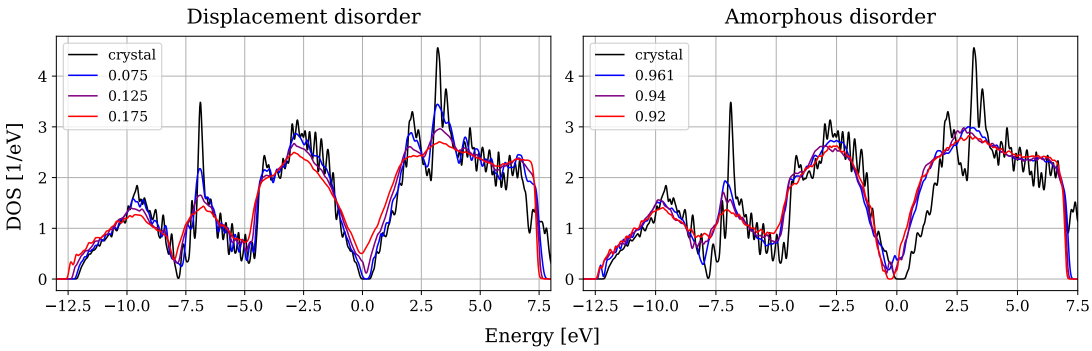

Figure 4.2: Density of states. The degree of displacement disorder is defined by the variance of random atomic displacements in Å, while amorphous disorder is characterized by the SOAP metric. Zero energy is set at the valence band maximum of the crystalline case.

Figure 4.2 shows the full valence and conduction bands for crystalline silicon and for increasing disorder. In the crystalline case, sharp van Hove singularities are visible, reflecting well-defined critical points in the band structure. As disorder increases, these features progressively broaden and lose their sharp structure. Such smoothing of van Hove singularities is a generic signature of disorder and has been extensively discussed in the theory of electronic states in non-crystalline materials [38, 39].

For displacement disorder, characterized by a root-mean-square atomic displacement in Å, the broadening occurs continuously. In contrast, for amorphous disorder quantified via the SOAP metric, the DOS undergoes more abrupt modifications. This difference reflects the fundamentally distinct structural nature of the two disorder classes: weakly perturbed periodic lattices versus topologically disordered networks.

It is worth noting that small oscillations visible in the crystalline DOS originate from finite k-point sampling in the reduced Brillouin zone of the supercell. Increasing k-point density reduces these numerical artifacts.

Figure 4.3 focuses on the region near the band gap. For displacement disorder, the gap decreases gradually as disorder increases. For amorphous disorder, however, the shift of the band edges occurs in a more step-like fashion. This reflects the absence of well-defined Bloch states and the formation of localized states near the mobility edge, a central concept in the theory of amorphous semiconductors [38].

Band-gap reduction with increasing disorder directly explains the red shift observed in the pump-induced current spectra in the following chapter. Since high-harmonic emission frequencies

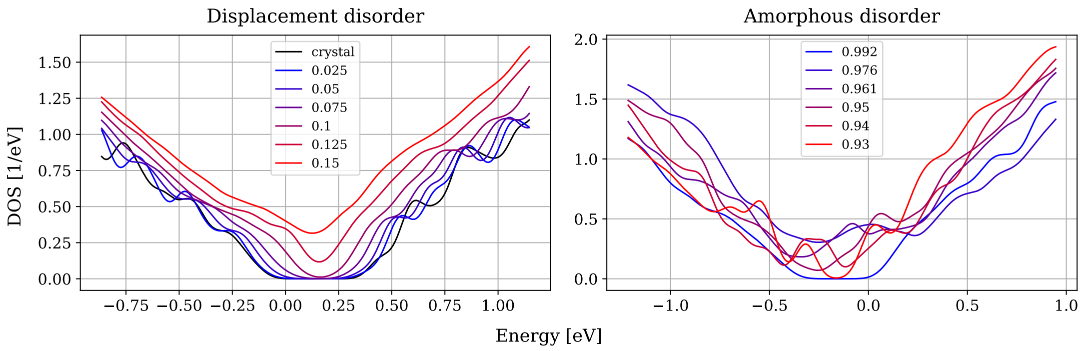

Figure 4.3: Density of states near the band gap for displacement and amorphous disorder.

scale with interband transition energies, disorder-induced gap narrowing shifts spectral weight to lower photon energies.

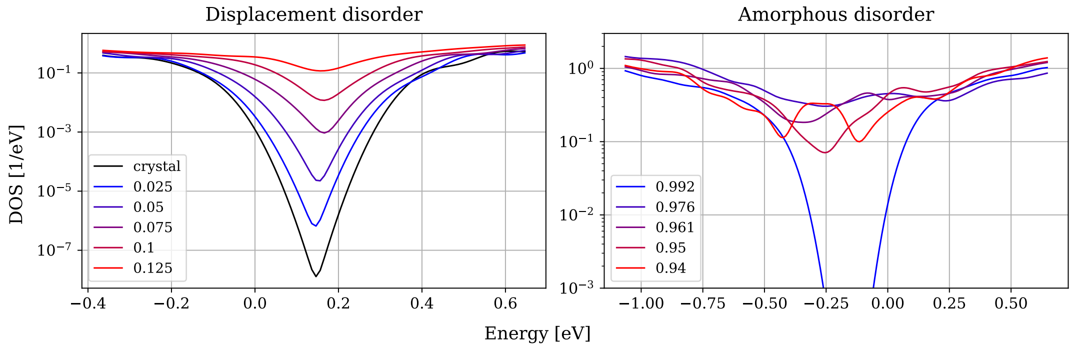

Figure 4.4: Logarithmic scale of the density of states near the band edge highlighting exponential tail formation.

In Figure 4.4, the DOS is plotted on a logarithmic scale to analyze the band-edge tails. In disordered semiconductors, the DOS near the band edge often follows the Urbach law,

$$g(E) \propto \exp\left(\frac{E - E_0}{E_U}\right),$$

where E_U is the Urbach energy [40]. The Urbach tail describes exponential decay of the DOS into the gap due to disorder-induced fluctuations of the potential landscape.

Although a clean exponential region is difficult to isolate in our finite supercells, the average slope of $\ln g(E)$ near the valence band edge decreases with disorder. This trend qualitatively correlates with the transport and dephasing times extracted later in this work. Physically, larger Urbach energies correspond to stronger spatial fluctuations of the local potential and therefore enhanced scattering.

From a localization perspective, exponential tails arise from spatially localized states in fluctuating potentials [39]. Such states contribute to enhanced inhomogeneous broadening and accelerate free-induction decay of coherent dipole oscillations.

Overall, the DOS analysis reveals three central disorder effects: First, progressive smoothing of van Hove singularities and reduction of band dispersion. Second, systematic band-gap narrowing and red shift of transition energies. Third, formation of exponential band-edge tails associated with disorder-induced localization.

These modifications of the electronic structure form the microscopic basis for the disorder dependence of photoinjection, current dynamics, and coherent oscillations discussed in subsequent chapters.

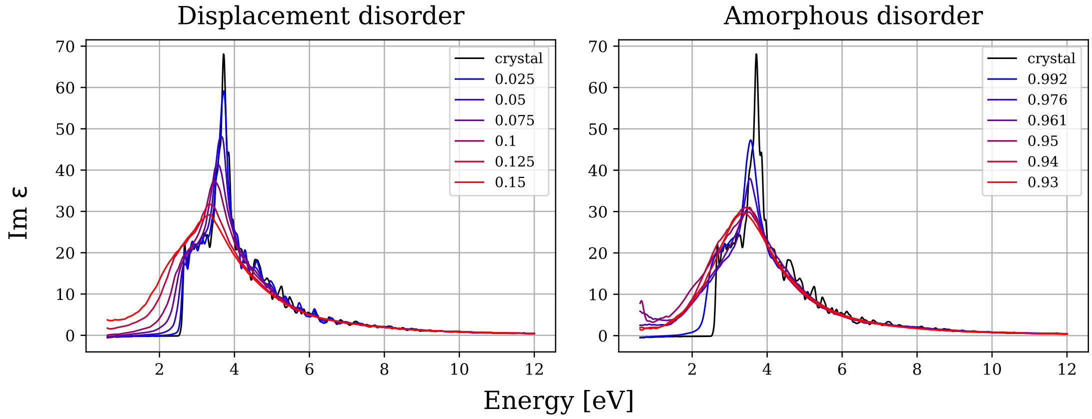

Figure 4.5: Imaginary part of the dielectric function.

Figure 4.5 shows the imaginary part of the complex dielectric function, $\varepsilon_2(\omega)$, which is directly associated with optical absorption and interband transitions. For the displacement disorder case, the direct band gap decreases systematically from 2.52 eV in the ideal crystal to 2.13 eV, 1.48 eV, 1.00 eV, and 0.60 eV as the displacement amplitude increases up to 0.1 Å. Beyond this point, the reduction becomes limited by the finite width of the time window used to compute the linear response, which in our case is 60 fs. In crystalline silicon, the absorption onset is sharp and reflects the well-defined band gap and van Hove singularities of the ordered band structure.

With increasing disorder, two systematic effects are observed. First, the absorption edge shifts to lower photon energies, indicating band-gap narrowing. This red shift directly mirrors the reduction of the gap seen in the density-of-states analysis (Fig. 4.3). Second, the absorption spectrum becomes progressively broadened and smoother. The sharp structures present in the crystalline case are washed out as disorder increases, consistent with the smoothing of van Hove singularities observed in the DOS (Fig. 4.2).

Physically, the broadening of $\varepsilon_2(\omega)$ reflects the increasing distribution of transition energies due to local potential fluctuations and reduced translational symmetry. The formation of band-tail states near the mobility edge, visible in the logarithmic DOS (Fig. 4.4), contributes to a gradual rather than abrupt onset of absorption. As a result, the energy at which absorption begins decreases with disorder, and the edge becomes less sharply defined.

Thus, the modifications of the imaginary part of the dielectric function provide an optical manifestation of the electronic-structure changes induced by disorder, linking band-gap reduction and DOS broadening directly to measurable absorption spectra.

4.3 Applicability of Strong Effective Disorder and Large Band-Gap Renormalization

In our RT-TDDFT supercell calculations, increasing structural disorder can reduce the Kohn–Sham band gap by up to $\sim$ 1 eV and even less to the point of disappearance, and produces pronounced broadening of the band-edge DOS and broadened absorption spectrum. At first sight, such large gap changes may appear inconsistent with experimentally realized amorphous silicon, where change in band gap is typically around 0.1 eV. This motivates a clarification of what the disorder parameter in the present work is meant to represent.

The central point is that the goal of this thesis is not to reproduce the long-time, equilibrium electronic gap of amorphous silicon. Rather, the disorder parameter is used as an *effective ultrafast scattering knob* that controls (i) the spectral width of interband transition energies contributing to macroscopic polarization and current, and (ii) the rate of elastic momentum randomization within the short ($\lesssim 30$ fs) time window relevant for attosecond pump–probe observables. In this regime, measured signals are primarily sensitive to coherence loss and transport relaxation rather than to the equilibrium definition of the optical gap. A clear example is provided by attosecond transient-absorption in crystalline Si: the electronic response to carrier injection builds up on sub-femtosecond time scales, while lattice-driven band-gap modifications appear only later, on $\sim 60 \pm 10$ fs time scales associated with the fastest optical phonon [2]. This separation of time scales justifies focusing on ultrafast electronic decoherence mechanisms when comparing to attosecond-scale measurements.

Homogeneous dephasing of an interband coherence $p_{\mathbf{k}}(t)$ is commonly modeled as

$$p_{\mathbf{k}}(t) \propto e^{-i\omega_{cv}(\mathbf{k})t}\, e^{-t/T_2}, \tag{4.1}$$

which corresponds in frequency space to a Lorentzian broadening of optical transitions with a characteristic width

$$\Gamma \sim \frac{\hbar}{T_2}. \tag{4.2}$$

Therefore, the femtosecond dephasing times frequently inferred in ultrafast solid-state spectroscopy correspond to very large effective broadenings. For example, $T_2 = 10$ fs implies $\Gamma \simeq 0.066$ eV, while $T_2 = 2$ fs implies $\Gamma \simeq 0.33$ eV, and $T_2 = 1$ fs implies $\Gamma \simeq 0.66$ eV. Broadening on the scale of 0.1–1 eV is thus the natural spectral signature of coherence decay in the experimentally relevant 1–10 fs range.

Ultrashort dephasing times of this order are not an artifact of our modeling: they are widely used and, in several settings, directly supported by experiments. In solid-state high-harmonic generation (HHG), reciprocal-space semiconductor-Bloch-equation simulations often require phenomenological dephasing times of only a few femtoseconds to reproduce measured spectra. Recent work has explicitly analyzed the *origin* of these "extremely short dephasing times" and argues that they are required to suppress contributions from large electron–hole separations during recombination [5]. Complementarily, phenomenological decoherence terms (e.g. imaginary potentials) have been introduced and validated against HHG trends, underscoring that femtosecond-scale dephasing is a key control parameter in solid HHG modeling [6]. Moreover, HHG-based metrology can access dephasing times directly: recent experiments used high-harmonic polarimetry to extract an electron–hole dephasing time T_2 of a few femtoseconds [4]. Although performed in a different material class, this result demonstrates that HHG observables can indeed be governed by femtosecond-scale coherence loss. Finally, even within silicon-specific ultrafast spectroscopy, phase-resolved transient absorption with sub-5-fs excitation reports an ultrafast single-electron momentum relaxation time of ~ 10 fs and Drude collision times down to a few femtoseconds at the earliest delays, together with a time-dependent optical effective mass [3]. This directly supports the physical relevance of the dephasing/transport window explored in the present simulations.

Within this perspective, the large reduction of the bandgap observed in our disordered supercells should be interpreted as a convenient proxy for accessing the regime of large effective broadening Γ and rapid dephasing (Eq. 4.2). In strongly disordered configurations, the notion of a single sharp Bloch band edge is replaced by broadened band-edge manifolds with tail states; consequently, optical transition energies relevant for current and polarization become distributed over a wide range.

We therefore view the disorder parameter primarily as an *effective ultrafast decoherence parameter* rather than a literal representation of equilibrium amorphous-silicon band-gap renormalization. This approach is appropriate for the thesis focus: linking disorder-induced DOS broadening, pump photoinjection, intraband current relaxation, and coherent-current-oscillation dephasing within a 0–30 fs window. At the same time, quantitative comparison to equilibrium optical gaps of a-Si:H would require realistic amorphous structural models (including hydrogen, defect statistics, and larger supercells) and, ultimately, quasiparticle corrections beyond Kohn–Sham eigenvalues [37]. Those extensions lie outside the present scope.

Chapter 5

Photoinjection

5.1 Pump and Probe Pulse Parameters

Understanding the transport dynamics of photoinjected carriers requires first establishing the excitation regime induced by the pump pulse. In this work, carrier generation is simulated within real-time time-dependent density functional theory (RT-TDDFT) using the SALMON code [12]. The calculations employ the PZ-LDA exchange-correlation functional [41], which, as is well known, underestimates semiconductor band gaps [37]. For crystalline silicon, according to the linear response analysis, SALMON shows a direct band gap of 2.52 eV., smaller than the real direct gap near 3.4 eV.

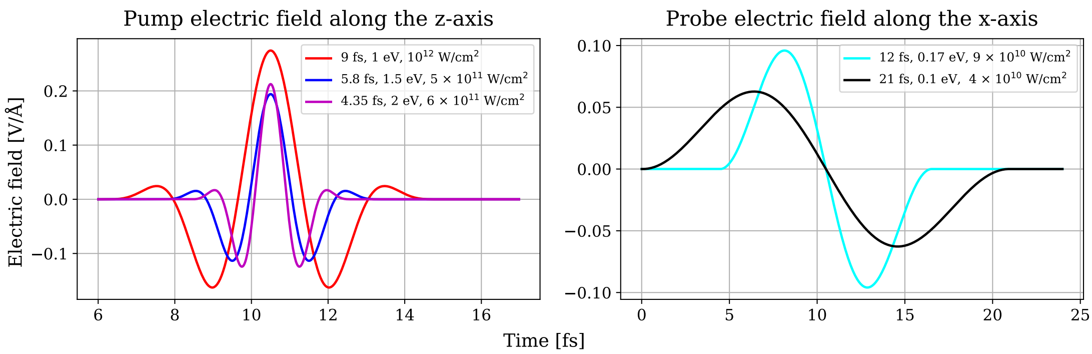

Figure 5.1: Temporal profiles of the pump (z-polarized) and probe (x-polarized) electric fields used in the simulations. The legend indicates pulse duration, carrier frequency, and peak intensity.

In this work, the pulses are introduced in the velocity gauge via the vector potential

$$A(t) = -\frac{E_0}{\omega}\cos^n\left(\frac{\pi}{T}\left(t-\frac{T}{2}\right)\right)\sin\left(\omega\left(t-\frac{T}{2}\right)+\varphi_{\mathrm{CEP}}\right), \qquad (0<t<T), \tag{5.1}$$

where φ_{CEP} is the carrier–envelope phase, T the pulse duration, ω the carrier frequency, and E_0 the electric-field amplitude. For the pump (probe) pulse we used $n = 4\,(2)$ and $\varphi_{\mathrm{CEP}} = 0\,(\frac{\pi}{2})$.

The corresponding electric field is obtained as

$$E(t) = -\frac{dA(t)}{dt}. \tag{5.2}$$

Figure 5.1 shows the electric fields of the pump and probe pulses. The pump pulse operates at carrier frequencies of 1 eV, 1.5 eV, and 2 eV. Because the TDDFT band gap is approximately 2.5 eV, excitation at 1 eV occurs in the multiphoton regime, at 1.5 eV in a reduced-order multiphoton regime, and at 2 eV close to a single-photon excitation threshold.

The nonlinear excitation mechanism in solids under strong fields has been extensively analyzed in the context of semiconductor Bloch equations and strong-field theory [42, 43, 44]. As the photon energy approaches the band gap, the effective order of the multiphoton process decreases and the excitation dynamics progressively approach linear absorption.

Disorder accelerates this transition to single-photon excitation. The reduction of the effective band gap and the breakdown of crystal momentum conservation allow additional transitions that are forbidden in a perfect lattice. Such relaxation of k-selection rules in disordered semiconductors is well established in optical absorption theory [38, 39].

It is important to emphasize that phonons are not included in the present RT-TDDFT simulations. Optical phonons in silicon have frequencies around 15 THz (period $\sim$ 65 fs), while acoustic phonons correspond to even longer time scales. Since we focus on dynamics within the first 30 fs after excitation, lattice motion can be neglected. On these ultrashort time scales, carrier dynamics are governed by purely electronic processes in a static potential.

The probe pulse is chosen such that it injects a negligible number of carriers. Even if weak photoinjection occurs, its contribution is removed by subtracting the probe-only response in pump–probe simulations.

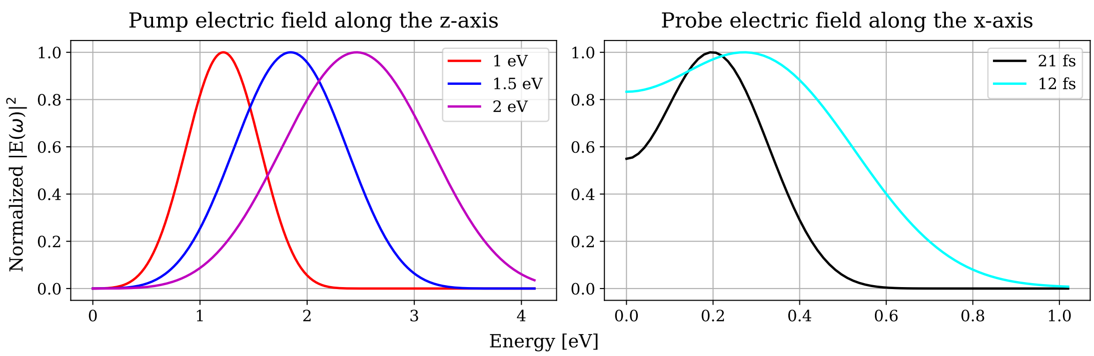

Figure 5.2: Power spectra of the pump and probe pulses.

Figure 5.2 shows the spectral intensities $|E(\omega)|^2$ of the pulses. Because the pulses are few-cycle in duration, their spectra are broadened according to the time–frequency uncertainty relation. Such broadening is characteristic of ultrashort strong-field excitation [45]. The spectral maxima exceed the nominal carrier frequencies due to the derivative in (5.2).

5.2 Pump Energy

We now analyze the electronic energy absorbed from the pump pulse and its dependence on disorder and carrier frequency.

Figure 5.3 shows that absorbed energy increases systematically with disorder. This increase reflects both band-gap narrowing and the relaxation of momentum conservation. Similar disorder-enhanced absorption has been reported in amorphous semiconductors and strongly perturbed crystals [38, 39].

As the pump photon energy increases from 1 eV to 2 eV, the curvature of the energy–time curve decreases, indicating a transition toward a more linear excitation regime. This behavior is consistent with strong-field solid-state theory, where higher photon energies reduce the effective nonlinearity of the excitation channel [43].

For amorphous disorder, the energy increase occurs stepwise rather than continuously. This behavior correlates with abrupt shifts in the density of states and band edges characteristic of amorphous silicon [38].

Figure 5.4 shows that the spread of absorbed energies decreases with increasing photon energy. In the single-photon regime (2 eV), disorder produces smaller relative variations in excitation efficiency.

Figure 5.5 quantifies statistical spread across independent supercells. The variance increases with disorder but decreases with photon energy, reflecting reduced sensitivity in the single-photon regime, σ denotes the overall relative variance of the spread of values.

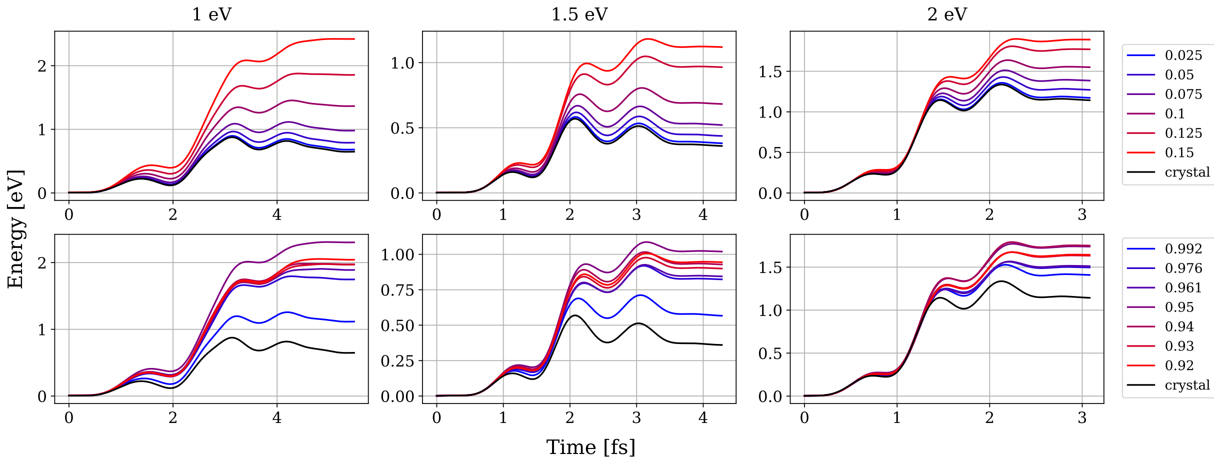

Figure 5.3: Time-dependent energy absorbed from the pump pulse for different photon energies and disorder strengths. The column caption in eV indicates the carrier frequency of the pump pulse. The top row represents displacement disorder, whereas the bottom row represents amorphous disorder.

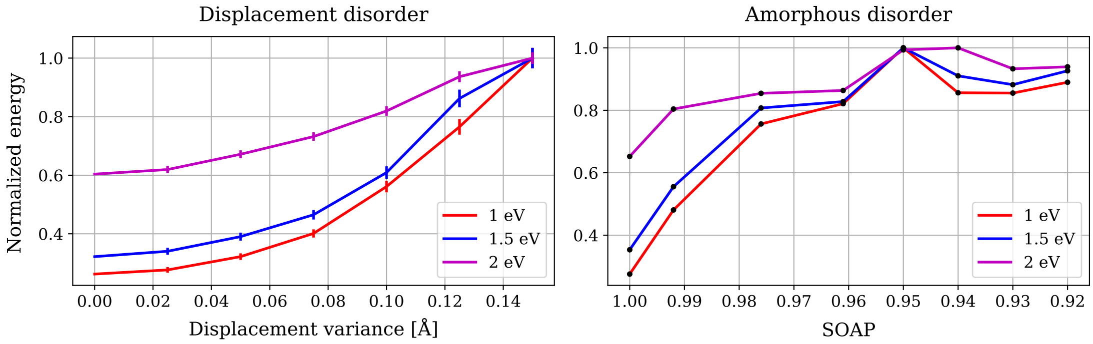

Figure 5.4: Final absorbed pump energy values normalized to the maximum value as a function of disorder strength.

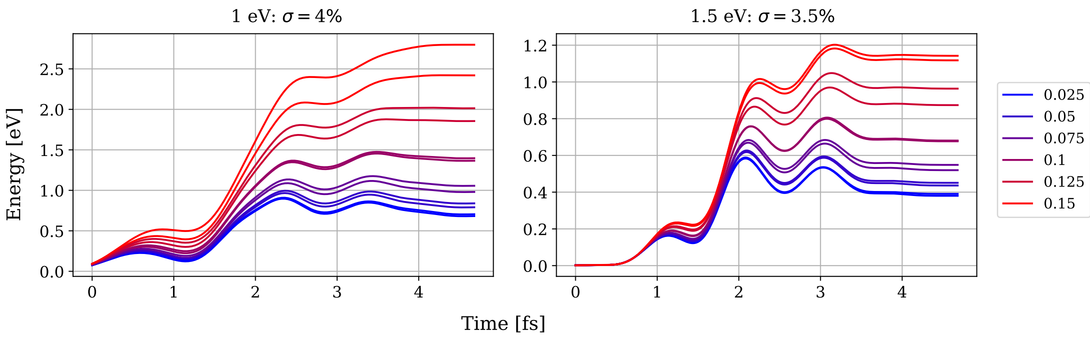

Figure 5.5: Statistical variation of absorbed pump energy across different displacement disorder realizations.

5.3 Pump-Induced Current

We now examine the current induced along the pump polarization direction.

Figure 5.6 shows increasing phase lag and suppression of coherent oscillations with disorder. After the pulse, oscillations decay more rapidly for stronger disorder, consistent with homogeneous

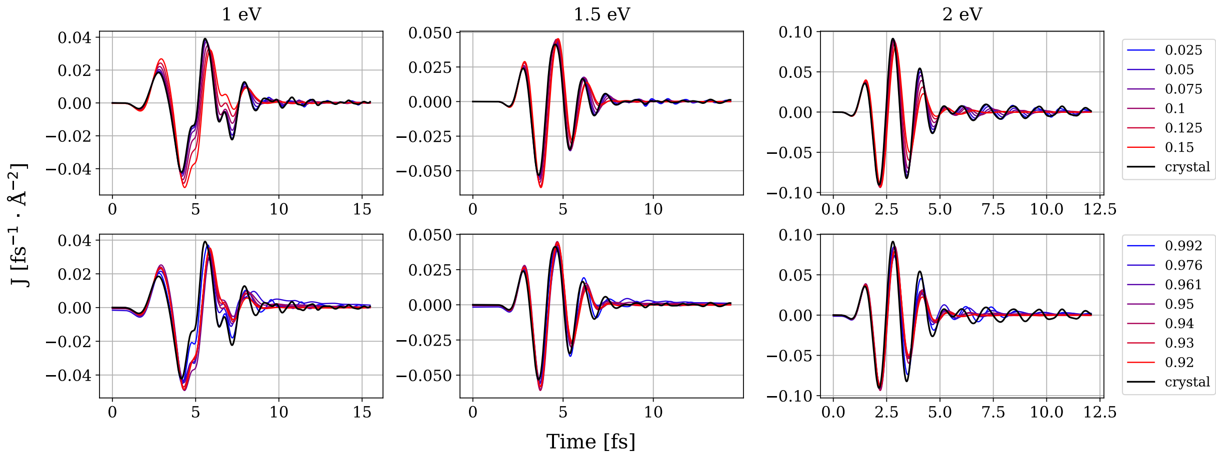

Figure 5.6: Time-domain current response induced by the pump pulse for increasing disorder. The column caption in eV indicates the carrier frequency of the pump pulse. The top row represents displacement disorder, whereas the bottom row represents amorphous disorder.

dephasing due to elastic scattering.

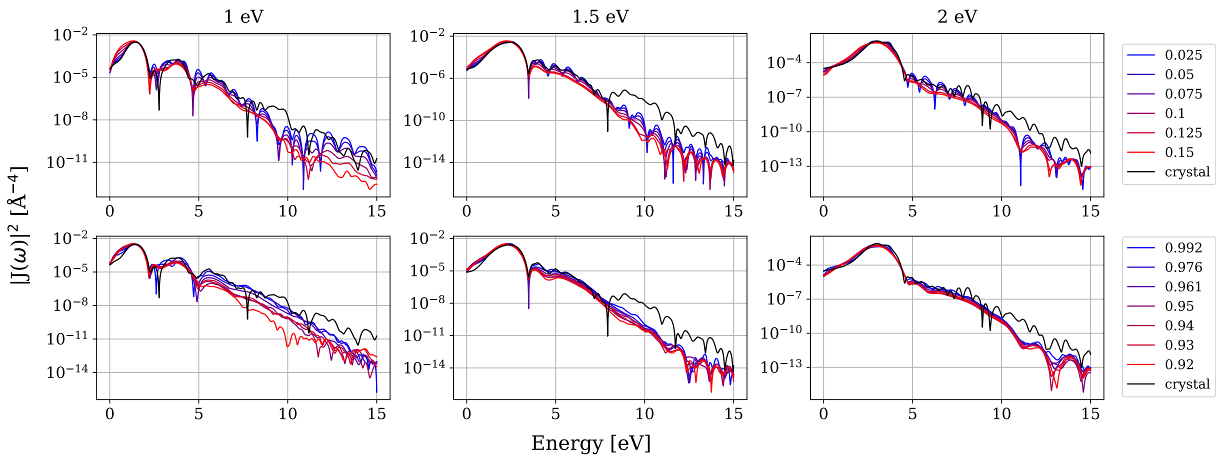

Figure 5.7: Logarithmic high-harmonic spectrum of the pump-induced current for different disorder strengths. The column caption in eV indicates the carrier frequency of the pump pulse. The top row represents displacement disorder, whereas the bottom row represents amorphous disorder.

The logarithmic spectrum in Figure 5.7 demonstrates suppression of high harmonics with increasing disorder. Disorder-induced reduction of high-harmonic generation efficiency has been observed in theoretical and experimental studies of HHG in solids [43, 44].

Figure 5.8 shows a clear red shift of the dominant spectral peak with increasing disorder, reflecting band-gap reduction. Spectral broadening accompanies the shift and can be attributed to relaxation of momentum conservation and activation of non-vertical transitions, a hallmark of disordered semiconductors [38, 39].

The spectral phase shown in Fig. 5.9 is obtained from the complex dielectric function

$$\varepsilon(\omega) = \varepsilon_1(\omega) + i\varepsilon_2(\omega), \tag{5.3}$$

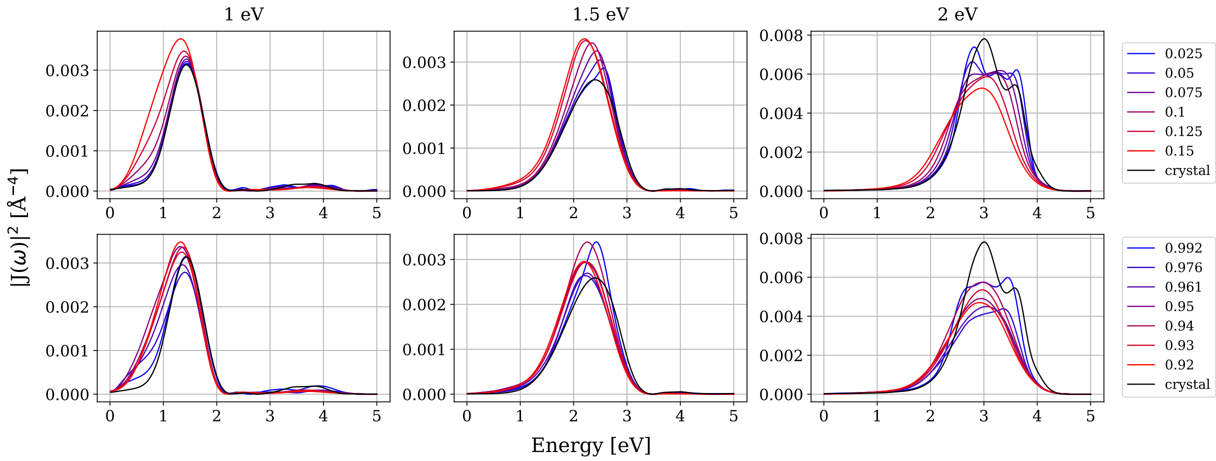

Figure 5.8: Linear-scale spectrum of the pump-induced current showing red shift and spectral broadening with disorder. The column caption in eV indicates the carrier frequency of the pump pulse. The top row represents displacement disorder, whereas the bottom row represents amorphous disorder.

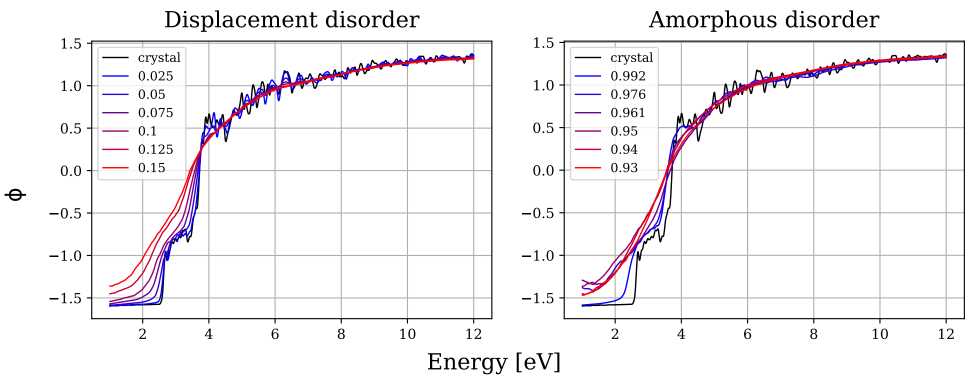

Figure 5.9: Spectral phase $\phi(\omega)$ extracted from the complex dielectric function $\varepsilon(\omega) = \varepsilon_1(\omega) + i\varepsilon_2(\omega)$ for displacement (left) and amorphous (right) disorder. In the energy range below $\sim$ 4 eV a clear increase of the phase with increasing disorder is observed, reflecting disorder-induced modification of the complex optical response.

via

$$\phi(\omega) = \arg \sigma(\omega) = \arctan\left(\frac{-[\varepsilon_1(\omega) - 1]}{\varepsilon_2(\omega)}\right). \tag{5.4}$$

As visible in Fig. 5.9, in the energy range below $\sim$ 4 eV the phase increases systematically with disorder strength for both displacement and amorphous configurations. This behavior reflects an enhanced phase lag of the optical response associated with the growing imaginary component of the dielectric function.

For a complete dynamical interpretation, however, it is not sufficient to consider $\phi(\omega)$ alone. One must also analyze the first and second spectral derivatives,

$$\frac{d\phi}{d\omega}, \qquad \frac{d^2\phi}{d\omega^2},$$

which determine the group delay and spectral dispersion, as well as the overall spectral weight distribution. Disorder-induced spectral broadening modifies the effective frequency content of the signal and therefore contributes to temporal reshaping beyond a simple uniform phase shift.

5.4 The Influence of the Probe Pulse on the Pump

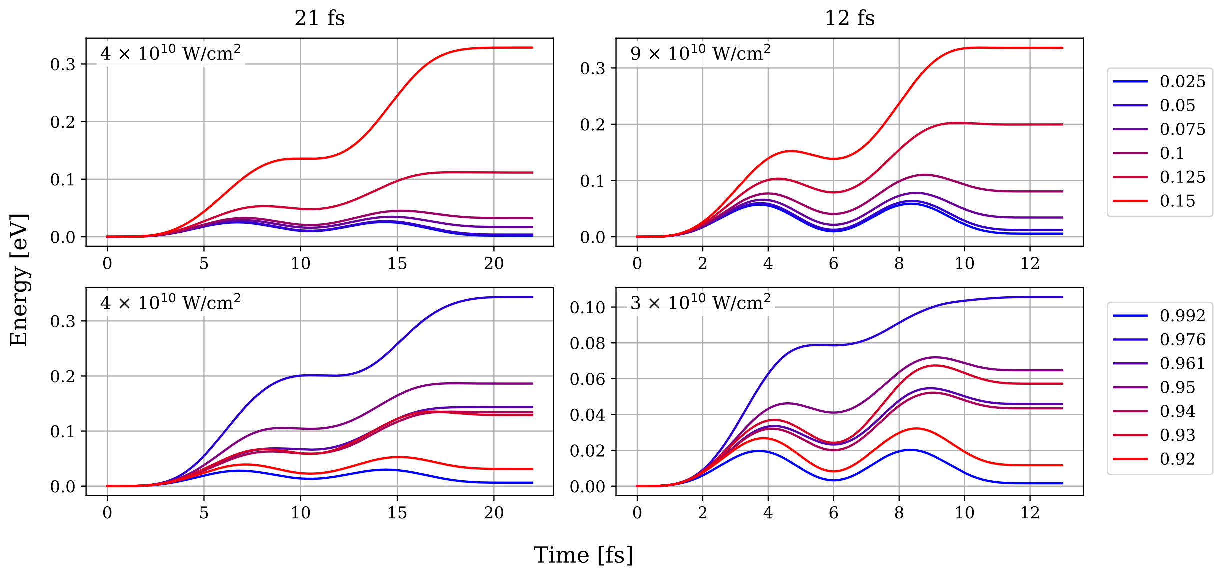

Figure 5.10: Electronic energy induced by probe pulses of different duration and intensity for various disorder strengths. The top row represents displacement disorder, whereas the bottom row represents amorphous disorder.

Figure 5.10 shows that for a 21 fs probe pulse with peak intensity 4×10^{10} W/cm^2, photoinjection is negligible across most disorder strengths. For extreme disorder, where the effective gap approaches the probe photon energy, weak injection becomes possible. Such behavior is expected when disorder significantly modifies the density of states and relaxes selection rules [38].

In the subsequent analysis, we therefore use the 21 fs probe pulse as the standard configuration. Any residual probe-induced excitation is subtracted in pump–probe simulations, ensuring that the measured orthogonal current reflects purely transport dynamics.

Chapter 6

Current Dynamics during the Probe Pulse

6.1 Pump-Probe Experiment Configuration

First, a pump pulse photoinjects charge carriers along the z-axis, and then a probe pulse shifts the electron wave packet along the x-axis. Orthogonally polarized pump–probe geometries are a standard strategy to separate carrier generation from transport, and closely related concepts are widely used in ultrafast and attosecond solid-state spectroscopy [2, 45]. In this section, we show that the probe-direction response can be described to a good extent by a Drude-type transport model with a disorder-dependent momentum relaxation time and an effective mass that changes as the wave packet is displaced in reciprocal space.

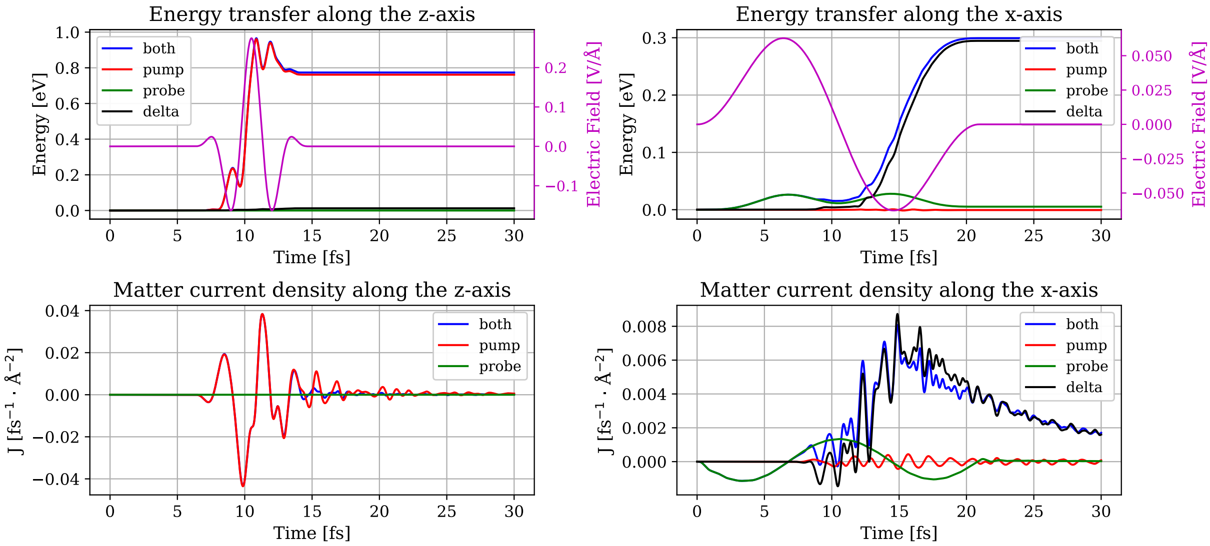

Figure 6.1: Example of pump-probe dynamics for 0.05 displacement disorder.

Figure 6.1 illustrates the pump–probe protocol and the decomposition of responses. In the legend, “both” denotes the simultaneous presence of the two pulses; “pump” denotes the pump-only response and “probe” denotes the probe-only response. We analyze energy transfer and matter current density along both polarization axes. The probe pulse is weak in the sense that its direct contribution to energy absorption during pumping is small; nevertheless, it measurably affects the pump-induced dynamics along z. A natural interpretation is that the orthogonal field perturbs the phase coherence of the interband polarization created by the pump, thereby modifying how efficiently the pump builds up coherent oscillators. Related pump–probe-induced modifications of coherence and band-structure response have been discussed in attosecond experiments in silicon [2].

Along the x-axis, the energy transfer remains negligible for small disorder, yet the probe-induced current can be large because it mainly reflects intraband acceleration. Therefore, to isolate transport of photoinjected carriers we analyze the current difference

$$\Delta J_x(t) = J_{\mathrm{both}}(t) - J_{\mathrm{pump}}(t) - J_{\mathrm{probe}}(t). \tag{6.1}$$

Subtracting the pump-only contribution is also essential because in a single finite supercell, local anisotropy can produce a small spurious orthogonal current. In a macroscopic disordered medium, or after sufficient ensemble averaging over disorder realizations, this term vanishes by symmetry.

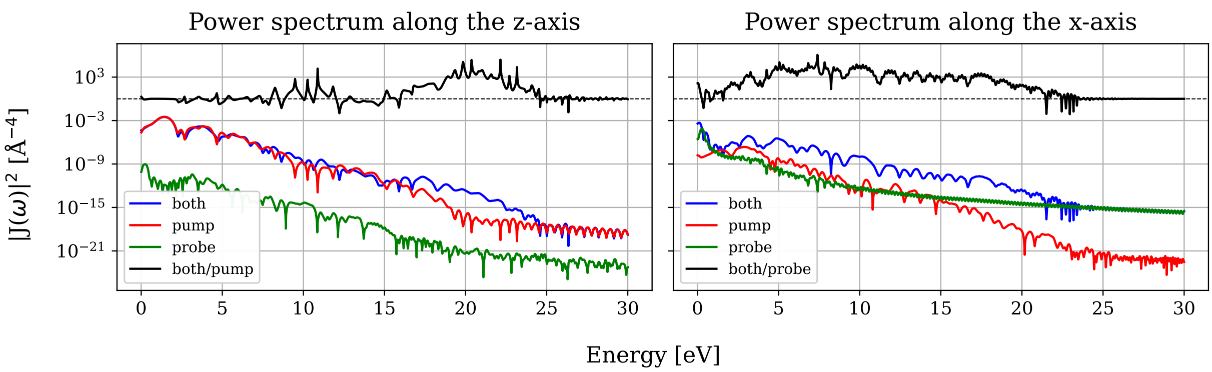

Figure 6.2: Power spectrum of currents. The ratio of the currents is shown in black.

The corresponding current power spectra are shown in Fig. 6.2. Along the z-axis, the probe present during pumping suppresses visibly coherent oscillations in the time domain, yet can increase nonlinearity signatures in the spectrum through field mixing. Along the x-axis, the pump–probe configuration enhances the low-frequency spectral weight compared with the probe-only case, consistent with the appearance of a long-lived current envelope arising from intraband motion of the photoinjected carrier ensemble. The separation between interband coherence (responsible for high-frequency components) and intraband transport (responsible for low-frequency components) is naturally formulated within semiconductor Bloch frameworks [42, 43].

6.2 Response to a Probe Pulse

We now focus on $\Delta J_x(t)$ as defined in Eq. (6.1) and study how disorder governs the probe-driven transport of photoinjected carriers.

Figure 6.3 shows that, after photoinjection, the probe generates a substantial current response which evolves during the probe field and decays afterward. In an ideal crystal within a collisionless RT-TDDFT setting, the post-pulse current does not decay because there is no intrinsic irreversible scattering channel. With static disorder, the current envelope decays, and as shown below this decay is well described by an exponential law. Such exponential relaxation is consistent with semiclassical elastic momentum scattering in a static disorder potential treated in the Born approximation [21, 46].

At the same time, the oscillatory structure sitting on top of the current envelope is suppressed with increasing disorder. These high-frequency oscillations are inherited from coherent interband dynamics excited by the pump; as disorder accelerates dephasing along z, the corresponding high-frequency components diminish and therefore become less visible in $\Delta J_x(t)$ as well.

Within the Drude picture, the probe-direction current obeys

$$\frac{dJ_x}{dt} = \frac{ne^2}{m^*(t)} E_x(t) - \frac{J_x}{\tau_m}, \tag{6.2}$$

where n is the photoinjected carrier density, τ_m is the momentum (transport) relaxation time, and $m^*(t)$ is an effective mass that may depend on the instantaneous reciprocal-space displacement

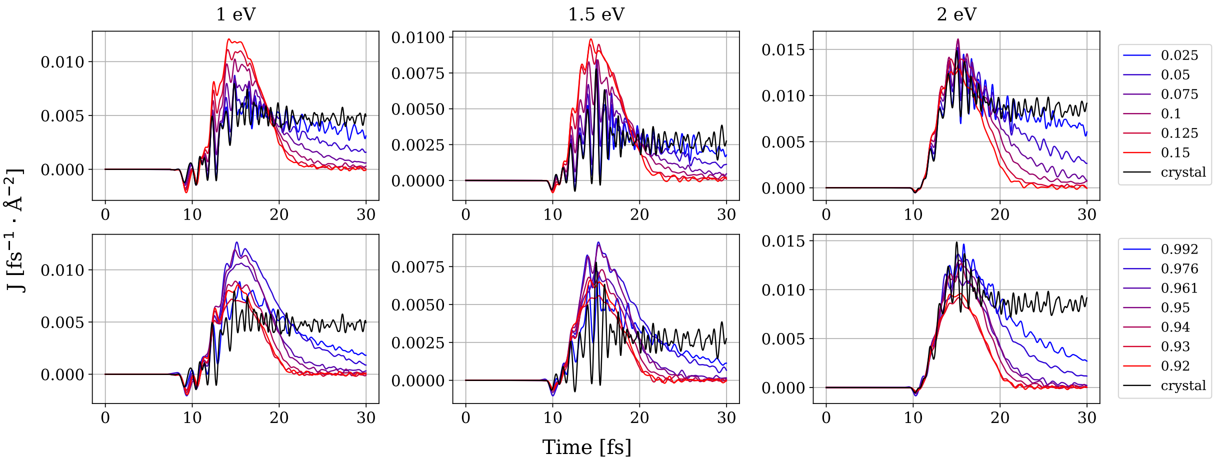

Figure 6.3: Pump-probe current response comparison. The column caption in eV indicates the carrier frequency of the pump pulse. The top row represents displacement disorder, whereas the bottom row represents amorphous disorder.

of the wave packet. The physically appropriate transport time is the momentum relaxation time, not the single-particle lifetime, and can be expressed as [46, 21]

$$\frac{1}{\tau_m(k)} = \sum_{k'} W_{k\to k'} \left(1 - \cos\theta_{kk'}\right), \tag{6.3}$$

where $W_{k\to k'}$ is the elastic scattering rate and $\theta_{kk'}$ is the scattering angle.

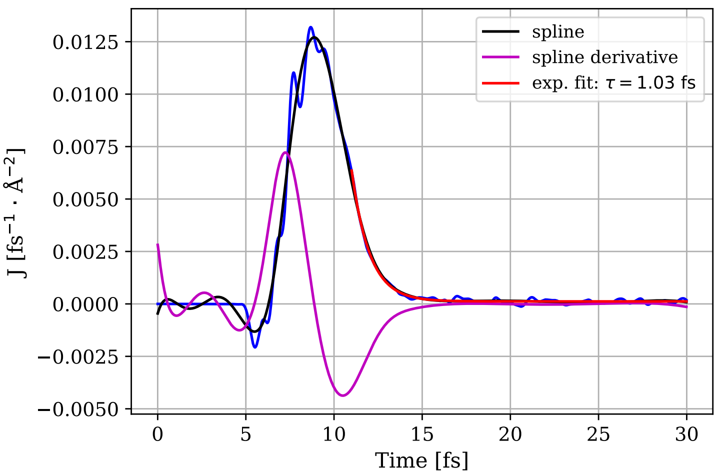

Figure 6.4: Example of a pump-probe response.

Figure 6.4 shows a representative pump–probe response and its envelope analysis. The tail is accurately described by an exponential function, allowing extraction of τ_m. We approximate the envelope by a spline (polynomial degree 5) and evaluate its derivative in the time window relevant to the second half-cycle of the probe pulse. Once τ_m is determined, Eq. (6.2) can be inverted to extract an effective mass from the measured derivative:

$$m^*(t) = \frac{ne^2 E_x(t)}{\frac{dJ_x}{dt} + \frac{J_x(t)}{\tau_m}}. \tag{6.4}$$

This procedure reveals that, while τ_m depends primarily on disorder strength, an accurate description of the waveform requires allowing m^* to vary appreciably as the wave packet is displaced in k-space.

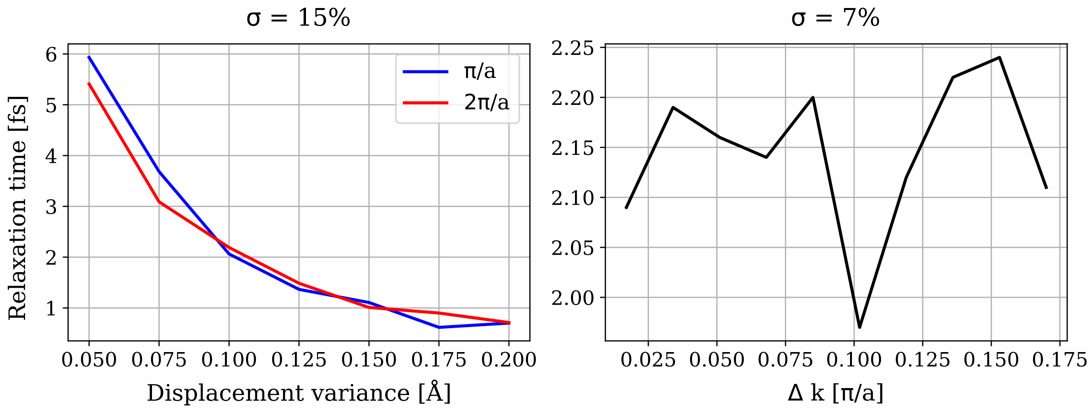

Figure 6.5: Momentum relaxation time dependence on crystal momentum.

Figure 6.5 demonstrates that the extracted transport time depends only weakly on the crystal-momentum shift induced by the probe pulse. The left panel compares two probe amplitudes corresponding to shifts of π/a and $2\pi/a$; despite the large difference in displacement, relative standard error is of about 15% overall. The right panel shows that, for a fixed disorder level (0.1 Å), τ_m has a relative standard error of about 7%. This supports treating τ_m as approximately constant for a given disorder strength and attributing the main remaining waveform variability to $m^*(t)$.

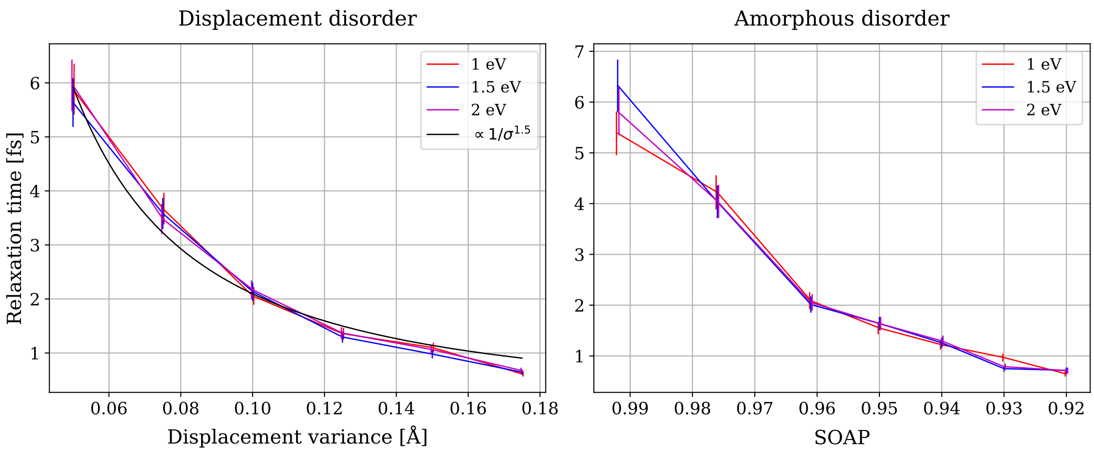

Figure 6.6: Momentum relaxation time dependence on disorder strength. The legend in eV denotes the carrier frequency of the pump pulse. For displacement disorder, a fit proportional to $\sigma^{-1.5}$ is shown for comparison with the black curve, where σ denotes the displacement variance.

Figure 6.6 shows that τ_m decreases monotonically with increasing disorder. Remarkably, for three different average pump photon energies (corresponding to different photoinjection regimes), the extracted transport times agree within the $\sim 10\%$ uncertainty. This indicates that momentum relaxation is controlled primarily by the static disorder potential, rather than by details of how the carriers were generated.

Figure 6.7 compares τ_m to the inverse Urbach energy extracted from the DOS tail analysis by averaging the derivative on a logarithmic scale and drawing a line according to the least squares

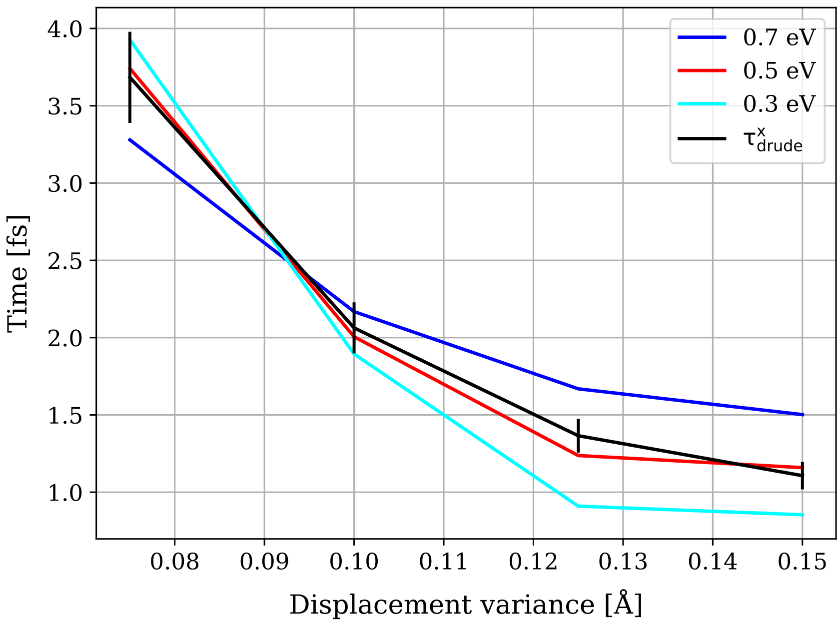

Figure 6.7: Urbach and relaxation time comparison.

method in the energy range indicated in the graph legend from the edge of the valence band and deep into it. In disordered semiconductors, exponential band-edge tails are commonly described by the Urbach rule [40, 38]. The observed correlation suggests that the same microscopic potential fluctuations that generate the band-edge tail also govern elastic scattering rates responsible for momentum relaxation. Although our finite-size DOS does not always provide an extended region of perfect exponential behavior, averaging the slope of $\ln g(E)$ over a controlled energy interval yields a robust proxy that tracks τ_m.

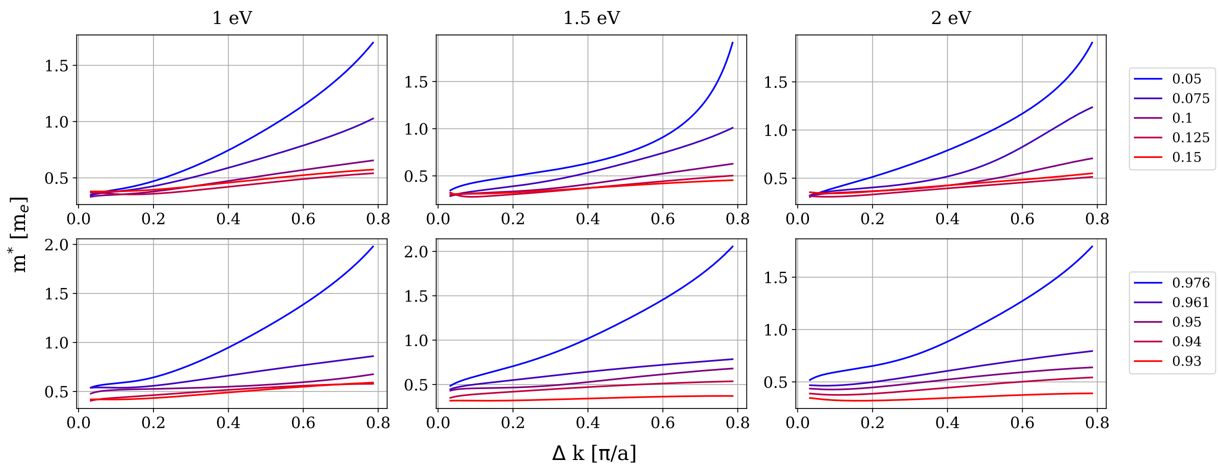

Figure 6.8: Crystal-momentum dependence of the effective mass. The column caption in eV indicates the carrier frequency of the pump pulse. The top row represents displacement disorder, whereas the bottom row represents amorphous disorder.

Figure 6.8 shows the extracted effective mass as a function of crystal-momentum displacement. The key trend is that with increasing disorder the range of m^* variation decreases. A consistent physical interpretation is that disorder relaxes momentum selection, enabling more uniform conduction-band population, which effectively averages over band-curvature variations and reduces the sensitivity of the wave-packet response to its instantaneous k-space position. Related dependence of effective mass on excitation regime and carrier distribution has been reported for ultrashort-pulse injection in solids [47].

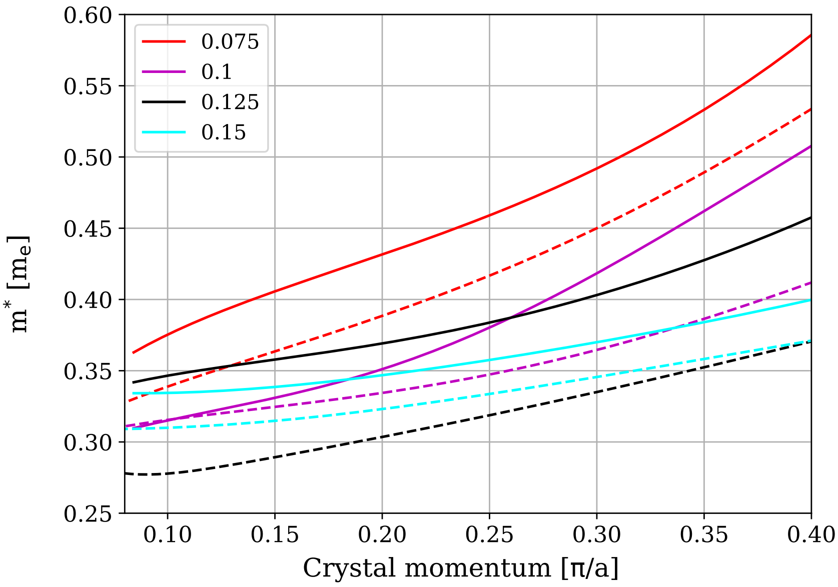

Figure 6.9: Comparison of the effective mass dependence for 12 fs and 21 fs probe pulses for different displacement disorders. The dashed(solid) line refers to the 12(21) fs pulse.

Finally, Fig. 6.9 validates the proposed extraction and modeling strategy. For different disorder realizations and two probe-pulse durations (12 fs and 21 fs), the effective-mass dependence inferred from Eq. (6.4) is consistent within the statistical uncertainty. The standard error is approximately 15% for m^*, while the spread for τ_m is about 10%. Since m^* varies by nearly a factor of two across the explored displacement range, this level of consistency provides strong evidence that a generalized Drude description with an approximately disorder-controlled τ_m and a k-dependent effective mass captures the essential physics of the probe-direction current dynamics.

Chapter 7

Coherent Current Oscillations

Before analyzing the oscillatory dynamics observed in the simulations, it is useful to clarify the terminology used in this chapter. In the present work we refer to the field-free oscillatory response of the electronic current as coherent current oscillations (CCO). Closely related behavior is widely known in spectroscopy under the term free induction decay (FID).

The concept of free induction decay was originally introduced in nuclear magnetic resonance (NMR), where a short excitation pulse creates a coherent macroscopic magnetization. After the driving field is removed, the magnetization continues to evolve freely and produces an oscillatory signal whose amplitude gradually decreases due to dephasing of the underlying microscopic degrees of freedom. An analogous phenomenon occurs in optical spectroscopy: an ultrashort laser pulse can create a coherent superposition of quantum states, giving rise to a macroscopic polarization that subsequently evolves in the absence of the external field and decays as phase coherence is lost.

In semiconductors excited by few-femtosecond optical pulses, the pump field generates interband coherence between valence- and conduction-band states. This coherence corresponds to a macroscopic polarization and is accompanied by an electronic current. After the excitation pulse has vanished, the coherent superposition continues to evolve freely with frequencies determined by the interband transition energies. As a result, the macroscopic current exhibits oscillations in the field-free regime. Their decay reflects the gradual loss of phase coherence due to two distinct mechanisms: homogeneous dephasing caused by scattering processes and inhomogeneous dephasing originating from the dispersion of transition energies across the Brillouin zone.

Although this behavior is formally analogous to optical free induction decay, the observable analyzed in this work is the electric current rather than the polarization. For this reason we introduce the term coherent current oscillations (CCO). The terminology emphasizes that the measured quantity in the simulations is the current response and that the oscillations arise from coherent evolving of electronic states. In this sense, CCO can be regarded as the current analogue of free induction decay: both phenomena originate from the free evolution of a coherently prepared electronic state, but CCO specifically refers to the oscillatory current signal generated by interband coherence in solids.

Adopting this terminology is particularly convenient in the present context. The pump–probe configuration used in the simulations naturally yields the time-dependent current as the primary observable, and its oscillatory component provides direct access to the coherence properties of the electronic system. As will be shown below, analyzing the CCO signal allows a quantitative separation of homogeneous and inhomogeneous dephasing mechanisms and establishes a direct connection between coherence decay and transport relaxation in disordered media.

7.1 Minimal Bloch-Basis Description: Semiconductor Bloch Equations

To connect the RT-TDDFT observables to a compact analytical picture, we employ the standard Bloch-basis density-matrix formulation of the semiconductor Bloch equations (SBE) [48, 49]. In a minimal two-band model (valence v, conduction c), for each crystal momentum $\mathbf{k}$ we introduce

$$f_c(\mathbf{k},t) = \rho_{cc}(\mathbf{k},t), \qquad f_v(\mathbf{k},t) = \rho_{vv}(\mathbf{k},t), \qquad p_{\mathbf{k}}(t) = \rho_{cv}(\mathbf{k},t), \tag{7.1}$$

where $\rho_{nm}(\mathbf{k},t)$ is the density matrix in the Bloch basis. The diagonal elements ρ_{cc} and ρ_{vv} correspond to the electron and hole populations, while the off-diagonal element ρ_{cv} describes the microscopic interband polarization.

The transition frequency is

$$\omega_{cv}(\mathbf{k}) = \frac{E_c(\mathbf{k}) - E_v(\mathbf{k})}{\hbar}, \tag{7.2}$$

where $E_{c(v)}(\mathbf{k})$ is the energy dispersion. The light–matter coupling can be written through the Rabi frequency

$$\Omega_{\mathbf{k}}(t) = \frac{1}{\hbar}\,\mathbf{E}(t)\cdot\mathbf{d}_{cv}(\mathbf{k}), \tag{7.3}$$

where the interband dipole matrix element is defined as

$$\mathbf{d}_{cv}(\mathbf{k}) = \langle u_{c\mathbf{k}}|e\cdot\mathbf{r}|u_{v\mathbf{k}}\rangle, \tag{7.4}$$

where $u_{c\mathbf{k}}$ and $u_{v\mathbf{k}}$ are the periodic parts of the Bloch functions corresponding to the conduction and valence bands, respectively.

In the absence of scattering, the SBE read [48, 49]

$$\dot{p}_{\mathbf{k}}(t) = -i\omega_{cv}(\mathbf{k})\,p_{\mathbf{k}}(t) - i\Omega_{\mathbf{k}}(t)\left[f_c(\mathbf{k},t) - f_v(\mathbf{k},t)\right], \tag{7.5}$$

$$\dot{f}_c(\mathbf{k},t) = 2\,\mathrm{Im}[\Omega^*_{\mathbf{k}}(t)\,p_{\mathbf{k}}(t)]\,, \qquad \dot{f}_v(\mathbf{k},t) = -\dot{f}_c(\mathbf{k},t). \tag{7.6}$$

After the pump pulse, $\Omega_{\mathbf{k}}(t) = 0$, and

$$\dot{f}_c(\mathbf{k}) = 0, \qquad \dot{p}_{\mathbf{k}} = -i\omega_{cv}(\mathbf{k})p_{\mathbf{k}} \tag{7.7}$$

which gives

$$p_{\mathbf{k}}(t) = p_{\mathbf{k}}(0)\,e^{-i\omega_{cv}(\mathbf{k})t}. \tag{7.8}$$

The parameters T_1 and T_2^* characterize two distinct relaxation processes of an excited electronic system. The time T_1 (population relaxation time) describes the decay of carrier populations toward equilibrium through processes such as recombination or energy relaxation. In contrast, total effective dephasing time T_2^* characterizes the decay of phase coherence between quantum states, i.e., the loss of the off-diagonal elements of the density matrix that determine the interband polarization.

Within the RT-TDDFT framework used in this work, no explicit dissipative channels such as carrier recombination, electron–phonon scattering, or other irreversible relaxation mechanisms are included. As a result, population relaxation is absent and one formally has $T_1 = \infty$. The dephasing time T_2^*, however, remains finite due to dephasing mechanisms. In a perfectly periodic system, the decay of the macroscopic polarization arises from inhomogeneous dephasing caused by the dispersion of interband transition frequencies $\omega_{cv}(\mathbf{k})$ across the Brillouin zone. In the presence of structural disorder, an additional mechanism contributes: elastic scattering of electronic states in both the valence and conduction bands, which leads to homogeneous dephasing and further accelerates the decay of coherence.

7.2 Interband Coherence and Inhomogeneous Dephasing

The macroscopic interband polarization is

$$P(t) = \sum_{\mathbf{k}} d_{cv}(\mathbf{k})\, p_{\mathbf{k}}(t), \tag{7.9}$$

and the interband current is $J_{\text{inter}}(t) = dP/dt$. After the pump pulse,

$$p_{\mathbf{k}}(t) = p_{\mathbf{k}}(0)\, e^{-i\omega_{cv}(\mathbf{k})t}. \tag{7.10}$$

Although each microscopic coherence is perfectly phase coherent, the macroscopic polarization

$$P(t) = \sum_{\mathbf{k}} A(\mathbf{k})\, e^{-i\omega_{cv}(\mathbf{k})t} \tag{7.11}$$

decays due to phase dispersion of $\omega_{cv}(\mathbf{k})$. For a Gaussian wave packet centered at $\mathbf{k}_0$ with width Δk,

$$P(t) \propto e^{-i\omega_0 t} \exp\left[-\frac{(\Delta\omega\, t)^2}{2}\right], \qquad \Delta\omega \simeq |\nabla_{\mathbf{k}}\omega_{cv}|\, \Delta k. \tag{7.12}$$

This defines the inhomogeneous dephasing time

$$T_{\text{inh}} = \frac{1}{\Delta\omega}. \tag{7.13}$$

In silicon, such dispersion-driven decay on the few-femtosecond scale is consistent with attosecond measurements of interband polarization dynamics [2, 50].

7.3 Static Disorder: Homogeneous Dephasing

When static disorder is introduced, elastic scattering modifies the SBE via collision integrals [48, 21]. In Born [46, 21] and effective-mass approximation [26],

$$W^{c(v)}_{\mathbf{k}\to\mathbf{k}'} = \frac{2\pi}{\hbar} n_i |\delta V^{c(v)}_{\mathbf{k}\mathbf{k}'}|^2 \delta(E_{c(v)}(\mathbf{k}) - E_{c(v)}(\mathbf{k}')), \tag{7.14}$$

where n_i is the concentration of the scattering centers, $|\delta V_{\mathbf{k}\mathbf{k}'}|^2$ is defined by 3.3, the indices c and v refer to the conduction and valence bands, respectively and $E_{c(v)}(\mathbf{k})$ is the energy dispersion. The single-particle scattering rate is

$$\Gamma_{c(v)}(\mathbf{k}) = \sum_{\mathbf{k}'} W^{c(v)}_{\mathbf{k}\to\mathbf{k}'}. \tag{7.15}$$

The interband dephasing time satisfies [51]

$$\frac{1}{T_2(\mathbf{k})} = \frac{1}{2}\big[\Gamma_c(\mathbf{k}) + \Gamma_v(\mathbf{k})\big]. \tag{7.16}$$

Thus, each microscopic coherence decays exponentially,

$$p_{\mathbf{k}}(t) = p_{\mathbf{k}}(0)\, e^{-i\omega_{cv}(\mathbf{k})t}\, e^{-t/T_2(\mathbf{k})}. \tag{7.17}$$

The macroscopic polarization envelope becomes approximately

$$P(t) \sim e^{-t/T_2} \exp\left[-\frac{t^2}{2T_{\text{inh}}^2}\right]. \tag{7.18}$$

7.4 Angular Factor and the Relation between T_2 and τ_m

Static disorder also relaxes intraband currents by randomizing momentum. However, the time scale governing current decay is not the same as the single-particle lifetime. The transport (momentum) relaxation time is defined by inserting an angular weight:

$$\frac{1}{\tau_m^{c(v)}(\mathbf{k})} = \sum_{\mathbf{k}'} W^{c(v)}_{\mathbf{k}\to\mathbf{k}'} \left(1 - \cos\theta_{\mathbf{k}\mathbf{k}'}\right), \tag{7.19}$$

where $\theta_{\mathbf{k}\mathbf{k}'}$ is the scattering angle. The factor $(1 - \cos\theta)$ encodes the fact that small-angle (forward) scattering barely changes the velocity direction.

It is useful to define the *angular factor*

$$\alpha_{c(v)}(\mathbf{k}) \equiv \frac{\sum_{\mathbf{k}'} W^{c(v)}_{\mathbf{k}\to\mathbf{k}'} \left(1 - \cos\theta_{\mathbf{k}\mathbf{k}'}\right)}{\sum_{\mathbf{k}'} W^{c(v)}_{\mathbf{k}\to\mathbf{k}'}}, \tag{7.20}$$

which satisfies $0 < \alpha_{c(v)}(\mathbf{k}) < 2$ and is typically $\alpha \sim 1$ for isotropic short-range disorder, while $\alpha \ll 1$ for forward-peaked scattering (e.g. screened Coulomb potentials). Combining Eqs. (7.15), (7.19), and (7.20) yields a direct relation between the single-particle rate and the transport time,

$$\Gamma_{c(v)}(\mathbf{k}) = \frac{1}{\alpha_{c(v)}(\mathbf{k})\, \tau_m^{c(v)}(\mathbf{k})}. \tag{7.21}$$

Substituting into Eq. (7.16) gives the connection we discussed:

$$\frac{1}{T_2(\mathbf{k})} = \frac{1}{2}\left[\frac{1}{\alpha_c(\mathbf{k})\, \tau_{m,c}(\mathbf{k})} + \frac{1}{\alpha_v(\mathbf{k})\, \tau_{m,v}(\mathbf{k})}\right]. \tag{7.22}$$

Two limits summarize the physics. If scattering is isotropic so that $\alpha \approx 1$ and if $\tau_{m,c} \approx \tau_{m,v}$ (which is true for the Born approximation and plane waves in the formula 3.3), then T_2 and τ_m are comparable. If scattering is strongly forward-peaked ($\alpha \ll 1$), then T_2 can be much shorter than τ_m: coherence decays rapidly even though the current persists relatively longer.

For a finite $\mathbf{k}$-distribution, the intraband current generally becomes a superposition of exponentials,

$$J_{\text{intra}}(t) = e \sum_{\mathbf{k}} v_c(\mathbf{k})\, \delta f_c(\mathbf{k}, 0)\, e^{-t/\tau_m(\mathbf{k})} + \cdots, \tag{7.23}$$

so a single τ_m may not describe the entire waveform. Nevertheless, Eq. (7.19) identifies which microscopic part of the impurity scattering controls the decay of the pump-induced drift/anisotropy.

7.5 Observed Coherent Current Oscillations Dynamics

Figure 7.1 shows the CCO along the z-axis after the pump pulse, i.e., in the absence of an external field. In a perfect crystal, the decay of the macroscopic signal is governed solely by inhomogeneous dephasing caused by dispersion of the transition frequency $\omega_{cv}(\mathbf{k})$ over the excited region of reciprocal space. As discussed above, this mechanism leads to an approximately Gaussian envelope characterized by the inhomogeneous time T_{inh}.

When static disorder is introduced, an additional homogeneous exponential decay channel appears due to elastic scattering. In that case, the macroscopic polarization and the corresponding interband current envelope can be written schematically as

$$J_z(t) \sim \exp\left(-\frac{t}{T_2}\right) \exp\left(-\frac{t^2}{2T_{\text{inh}}^2}\right), \tag{7.24}$$

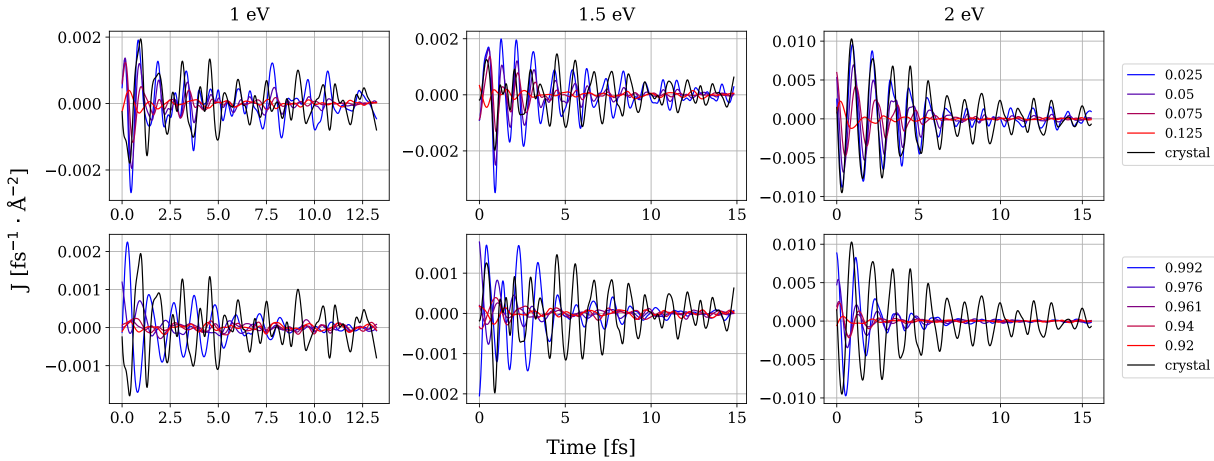

Figure 7.1: Field-free coherent current oscillations $J_z(t)$ after the pump pulse for different disorder strengths and pump photon energies. The column caption in eV indicates the carrier frequency of the pump pulse. The top row represents displacement disorder, whereas the bottom row represents amorphous disorder.

where T_2 is the homogeneous dephasing time associated with impurity scattering, while T_{inh} describes dispersion-driven phase mixing. The total effective dephasing time T_2^* is therefore determined by both mechanisms, and in the regime where the exponential decay dominates one may approximate

$$\frac{1}{T_2^*} \equiv \frac{1}{\tau_{\mathrm{coh}}^z} \approx \frac{1}{T_2} + \frac{1}{T_{\mathrm{inh}}^{\mathrm{eff}}}. \tag{7.25}$$

Although in the case where we have several spectral peaks, the attenuation shape is not Gaussian, such a division into different dephasing channels will still be justified, since exponential attenuation is uniform.

It is clearly seen in Fig. 7.1 that the oscillations decay more rapidly as disorder increases, indicating a progressive reduction of T_2. Moreover, in the pumping regime closest to linear absorption (carrier frequency 2 eV), the oscillations are spectrally cleaner and exhibit a more regular phase evolution. In contrast, strongly nonlinear pumping (1 eV and 1.5 eV) generates a broader and more complex distribution of microscopic phases, which enhances inhomogeneous contributions and leads to a more irregular macroscopic waveform.

The corresponding spectra in Fig. 7.2 show a systematic red shift with increasing disorder, consistent with band-gap reduction. Spectral broadening is also evident. For amorphous disorder under nonlinear pumping, the spectrum becomes significantly more irregular compared to displacement disorder, indicating abrupt activation of transitions that are symmetry-forbidden in the crystalline limit. For nearly linear pumping at 2 eV, the spectral modifications are much weaker, reflecting the more selective excitation of states near the band edge.

7.6 Dephasing Times Decomposition

Figure 7.3 shows a representative CCO waveform for a displacement disorder of 0.05 Å. Our goal is to separate the homogeneous exponential decay from the residual inhomogeneous contribution and to test whether the two mechanisms can be decomposed additively at the level of decay rates.

To isolate the non-exponential component, we normalize the signal by the homogeneous decay factor associated with the transport time τ_m obtained from the probe-direction Drude analysis:

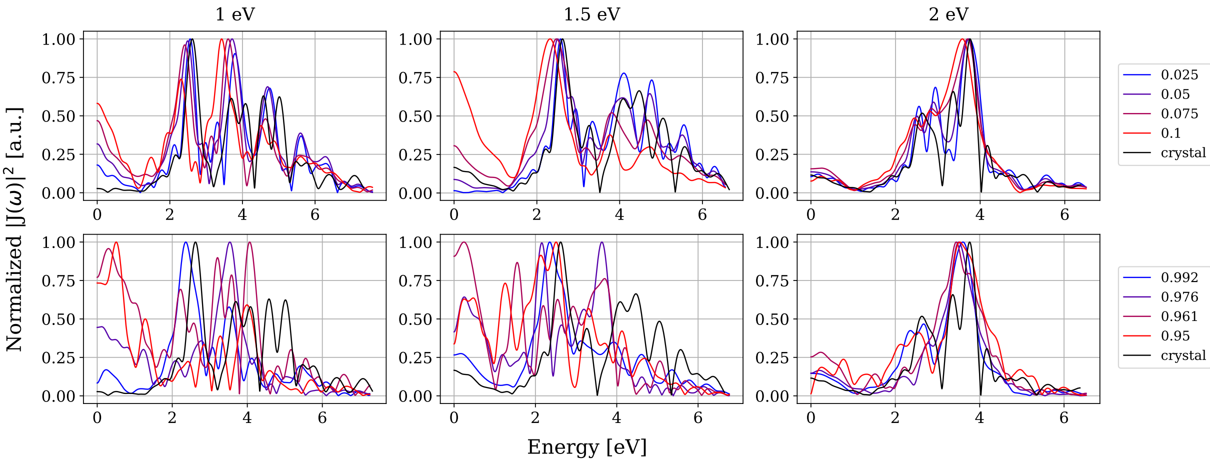

Figure 7.2: Power spectrum of coherent current oscillations $|J_z(\omega)|^2$ for increasing disorder. The column caption in eV indicates the carrier frequency of the pump pulse. The top row represents displacement disorder, whereas the bottom row represents amorphous disorder.

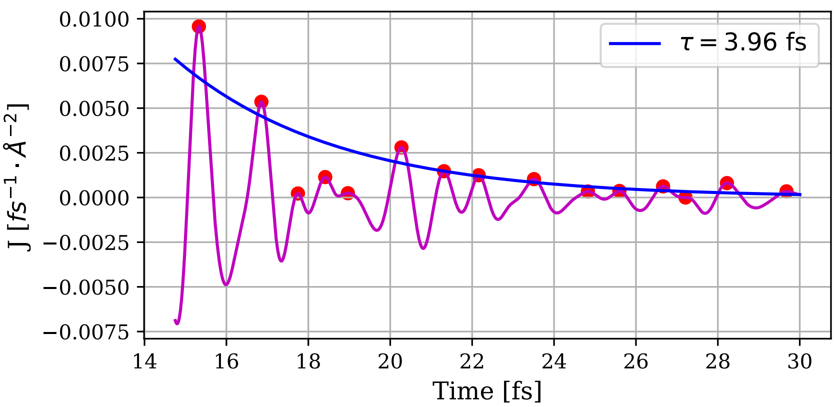

Figure 7.3: Example of a coherent current oscillation signal used for coherence-time analysis.

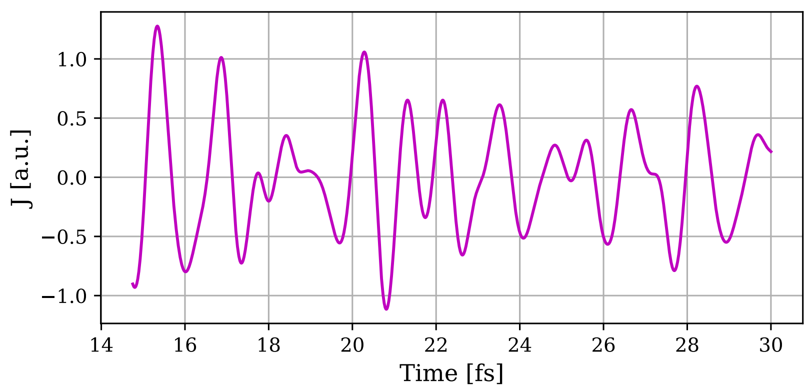

Figure 7.4: Exponentially normalized coherent current oscillation signal.

$$\tilde{J}_z(t) = J_z(t)\, e^{(t-t^*)/\tau_m}. \tag{7.26}$$

The normalized signal $\tilde{J}_z(t)$, shown in Fig. 7.4, should primarily contain inhomogeneous phase mixing if the homogeneous part is indeed governed by the same scattering time τ_m.

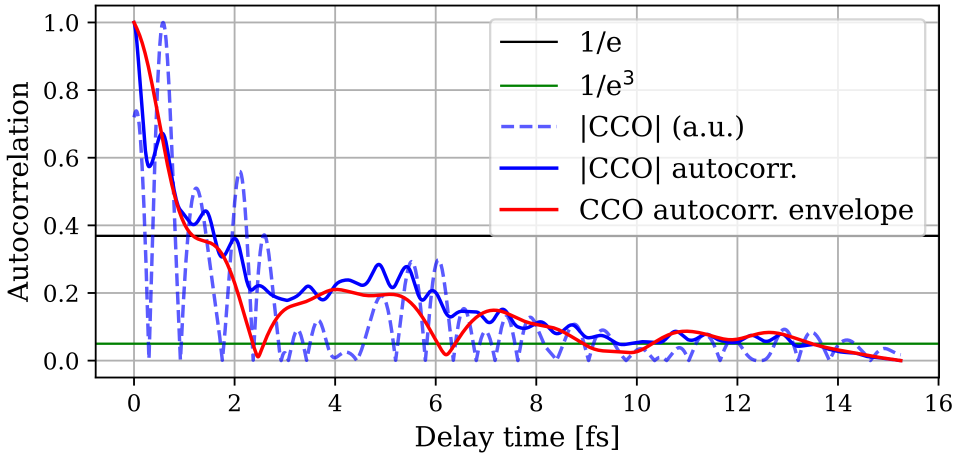

Figure 7.5: Autocorrelation of the CCO signal and its Hilbert-envelope used for coherence-time extraction.

To quantify coherence, we compute the autocorrelation of the absolute signal,

$$C(\tau) = \int_{t_0}^{\infty} |J_z(t)|\,|J_z(t+\tau)|\,dt, \qquad \tau \geq 0. \tag{7.27}$$

We then construct the analytic signal via the Hilbert transform $\mathcal{H}$:

$$\mathcal{C}(\tau) = C(\tau) + i\,\mathcal{H}[C](\tau), \tag{7.28}$$

and define the envelope

$$A(\tau) = |\mathcal{C}(\tau)|. \tag{7.29}$$

Following coherence theory concepts [52], we define the coherence decay time as

$$\tau_{\text{coh}} \equiv 2 \int_0^{\infty} \left|\frac{A(\tau)}{A(0)}\right|^2 d\tau \tag{7.30}$$

The prefactor 2 accounts for restricting the analysis to positive delays.

Figure 7.6 compares various coherence-time definitions. The most relevant quantity is τ^z_{coh}, representing the total dephasing time of the field-free CCO signal. Also shown is the coherence decay time extracted from the exponentially normalized signal, isolating inhomogeneous effects. The transport time τ_m is plotted for reference. Remarkably, the inverse spectral width (obtained from the squared autocorrelation integral) correlates strongly with τ^z_{coh}, indicating that spectral broadening and temporal dephasing are directly linked. The coherence decay time determined at the $1/e$ level for the autocorrelation of the signal module gives a good similarity to τ^z_{coh}. What is also evident is that the definition at the $1/e^3$ level allows us to approximately extract a longer-range correlation due to exponential decay for displacement variances below 0.08 (while exponential decay takes longer than inhomogeneous decay). The comparison suggests the presence of two characteristic time scales: a shorter time arising from inhomogeneous broadening and a longer time reflecting homogeneous scattering.

Finally, Fig. 7.7 tests the relation

$$\frac{1}{\tau^z_{\text{coh}}} = \frac{1}{\tau_m} + \frac{1}{\tau^{z,\text{norm}}_{\text{coh}}}. \tag{7.31}$$

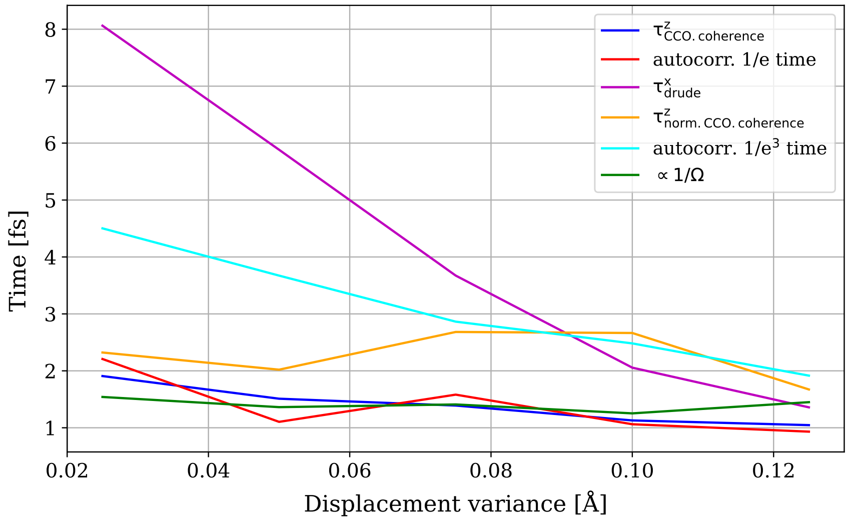

Figure 7.6: Comparison of different coherence-time definitions and their relation to the transport time.

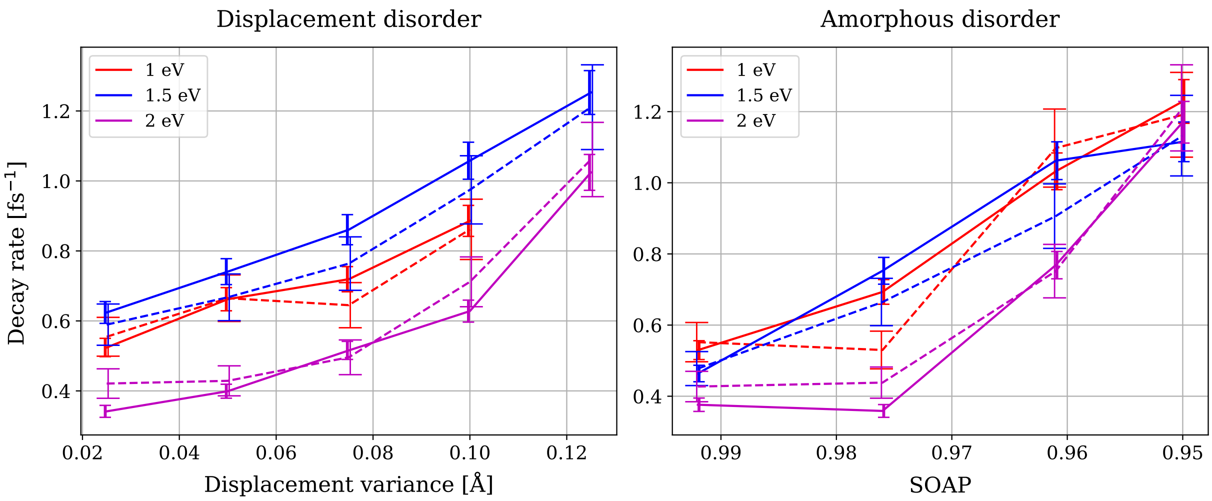

Figure 7.7: Test of additivity of decay rates: comparison of $1/\tau^{z}_{\mathrm{coh}}$ with $1/\tau_m + 1/\tau^{z,\mathrm{norm}}_{\mathrm{coh}}$.

Here $\tau^{z,\mathrm{norm}}_{\mathrm{coh}}$ is the coherence decay time for the normalized CCO signal. For nearly all points, the error bars overlap. This shows that the homogeneous part of CCO decay is governed by the same elastic scattering time that controls intraband transport. Consequently, the effective angular factor in the transport time definition is close to unity, indicating nearly isotropic impurity scattering in the bias-disorder model considered here.

Chapter 8

Conclusion

The present work provides a unified picture of ultrafast electronic dynamics in disordered silicon, combining real-time TDDFT simulations with analysis of expressions from the semiconductor Bloch equations and Drude transport theory. By systematically varying both structural disorder and excitation regime, we have identified how disorder modifies (i) the electronic structure, (ii) photoinjection efficiency, (iii) intraband transport, and (iv) coherent interband dynamics.

Electronic Structure as the Microscopic Origin

The density-of-states analysis establishes the microscopic foundation for all subsequent dynamical effects. With increasing disorder, three key modifications occur: smoothing of van Hove singularities, systematic band-gap reduction, and formation of near-exponential band-edge tails consistent with Urbach-type behavior [38, 40].

Band-gap narrowing explains the red shift observed in high-harmonic spectra and coherent current oscillation (CCO) spectra. At the same time, the emergence of tail states increases the density of available transition frequencies near the band edge, thereby enhancing inhomogeneous dephasing. Importantly, the Urbach energy extracted from the DOS correlates with the momentum relaxation time obtained from transport analysis, indicating that both spectral broadening and carrier scattering originate from the same disorder-induced potential fluctuations.

Photoinjection Regimes and Nonlinearity

The photoinjection analysis demonstrates that excitation regime plays a central role in shaping subsequent dynamics. When the pump photon energy approaches the effective band gap, excitation enters a single-photon regime. In this limit, photoinjection becomes less sensitive to disorder-induced variations in local structure, and spectral phase distributions remain relatively narrow.

In contrast, multiphoton excitation produces broader distributions in $\mathbf{k}$-space. This leads to stronger inhomogeneous dephasing even in the absence of scattering. Disorder accelerates the transition from multiphoton-dominated excitation toward effectively linear absorption by relaxing momentum selection rules, consistent with general disorder-induced breakdown of Bloch momentum conservation [39].

Transport: Momentum Relaxation and Effective Mass Renormalization

The probe-direction current analysis reveals that intraband transport is well described by a generalized Drude model with a disorder-controlled momentum relaxation time τ_m and a $\mathbf{k}$-displacement-dependent effective mass $m^*(k)$.

The key observation is that τ_m depends primarily on disorder strength and only weakly on pump photon energy or probe-induced $\mathbf{k}$ displacement. The extracted relaxation times decrease monotonically with disorder, reflecting increased scattering probability due to stronger potential fluctuations.

At the same time, the effective mass exhibits significant variation with reciprocal-space displacement, especially in the low-disorder regime. As disorder increases, the range of m^* variation decreases. Physically, this can be understood as disorder-induced averaging over band curvature due to relaxation of strict momentum selection, similar to trends observed in strong-field excitation [47].

Coherent Current Oscillations: Dephasing Mechanisms

The CCO analysis provides the clearest separation between homogeneous and inhomogeneous dephasing mechanisms. In the absence of disorder, decay is governed entirely by phase mixing across the excited $\mathbf{k}$ region, yielding an inhomogeneous dephasing time T_{inh}. No microscopic irreversible process is present in RT-TDDFT, and therefore $T_2 = \infty$ in the absence of disorder.

With static disorder, elastic scattering introduces a finite homogeneous dephasing time T_2. The observed envelope of coherent current oscillations is well described by a product of exponential (homogeneous) and Gaussian-like (inhomogeneous) contributions.

A central result of this work is that the homogeneous component of CCO decay is quantitatively consistent with the transport relaxation time extracted independently from probe-direction dynamics. Within error bars, for almost all points the decay-rate additivity relation

$$\frac{1}{\tau_{\text{coh}}^{z}} = \frac{1}{\tau_m} + \frac{1}{\tau_{\text{coh}}^{z,\text{norm}}} \tag{8.1}$$

is satisfied. This implies that the scattering responsible for momentum relaxation also governs interband coherence decay. In terms of scattering theory, this indicates that the angular factor in the transport-time definition is close to unity, corresponding to nearly isotropic impurity scattering in the displacement-disorder model.

Unified Physical Picture

Taken together, the results support a consistent physical interpretation:

1. Structural disorder reshapes the electronic density of states, producing band-gap reduction and near-exponential band-edge tails.
2. These structural modifications determine both photoinjection efficiency and the spectral distribution of excited states.
3. Elastic scattering from the same disorder potential governs both momentum relaxation and homogeneous coherence decay.
4. Inhomogeneous dephasing arises from dispersion of transition frequencies across the excited $\mathbf{k}$ region and is enhanced by nonlinear excitation.
5. Intraband transport can be accurately described by a generalized Drude model, incorporating a disorder-induced momentum relaxation time τ_m and an effective mass $m^*(k)$ that depends on the $\mathbf{k}$-space displacement.

Thus, transport time, homogeneous dephasing time, and Urbach energy are not independent quantities but manifestations of the same underlying disorder potential.

Implications

The demonstrated equivalence between the homogeneous component of CCO decay and the transport relaxation time suggests that ultrafast coherent-current measurements can provide direct access to transport properties on femtosecond time scales. This connects attosecond solid-state spectroscopy [2, 50] with semiclassical transport theory in a quantitative manner.

Moreover, the weak dependence of τ_m on excitation regime indicates that ultrafast transport properties in disordered silicon are controlled predominantly by structural disorder rather than by strong-field excitation pathway. This robustness enhances the interpretability of pump–probe experiments in realistic materials.

Outlook

The present analysis neglects phonons and electron–electron scattering, restricting attention to sub-30-fs dynamics. On longer time scales, additional irreversible channels would further reduce coherence and modify transport. Incorporating lattice motion within time-dependent frameworks would allow exploration of the crossover between purely electronic dephasing and phonon-assisted relaxation.

Nevertheless, within the ultrafast window considered here, the combined TDDFT–Drude–SBE framework provides a consistent and quantitatively supported description of how structural disorder governs electronic dynamics in silicon.

Appendix A

Density of States with Scattering Broadening

A.1 Analytical Calculation of the Integral for Scattering Broadening

Evaluation of the broadened DOS integral

We consider the integral

$$I(u,a) = \int_{-\infty}^{+\infty} \frac{t^2}{(t^2-u)^2+a^2}\,dt, \qquad a>0, \tag{A.1}$$

which appears in the calculation of the disorder-broadened DOS within the effective-mass approximation.

Step 1: Complex-analytic representation. Define the complex function

$$f(z) = \frac{z^2}{(z^2-u)^2+a^2}, \tag{A.2}$$

and consider the contour integral of $f(z)$ along the real axis closed by a semicircle in the upper half-plane. For $|z| \to \infty$, $f(z) \sim 1/z^2$, so the contribution from the semicircle vanishes, and we have

$$\int_{-\infty}^{\infty} f(t)\,dt = 2\pi i \sum_{z_j \in \mathrm{UHP}} \operatorname*{Res}_{z=z_j} f(z), \tag{A.3}$$

where the sum runs over the poles in the upper half-plane (UHP).

The poles of $f(z)$ are determined by

$$(z^2-u)^2+a^2 = 0 \;\;\Rightarrow\;\; z^2-u = \pm ia \;\;\Rightarrow\;\; z^2 = u \pm ia. \tag{A.4}$$

Thus there are four simple poles,

$$z = \pm z_1, \quad z = \pm z_2, \qquad z_1^2 = u+ia, \quad z_2^2 = u-ia. \tag{A.5}$$

We choose the standard branches of the square roots so that z_1 and z_2 lie in the upper half-plane (i.e. $\Im z_{1,2} > 0$). These two poles contribute to the contour integral.

Step 2: Residues at the poles. We write the denominator as

$$(z^2-u)^2+a^2 = (z^2-u-ia)(z^2-u+ia) \equiv D_1(z)D_2(z).$$

At a simple pole z_0 where $D_1(z_0)=0$ and $D_2(z_0)\neq 0$, the residue is

$$\operatorname*{Res}_{z=z_0} f(z) = \frac{z_0^2}{D_1'(z_0)D_2(z_0)}.$$

Pole at $z=z_1$ (where $z_1^2 = u+ia$, i.e. $D_1(z) = z^2-u-ia$):

$$D_1'(z) = 2z, \qquad D_2(z_1) = z_1^2 - u + ia = (u+ia) - u + ia = 2ia. \tag{A.6}$$

Hence

$$\operatorname*{Res}_{z=z_1} f(z) = \frac{z_1^2}{(2z_1)(2ia)} = \frac{z_1}{4ia}. \tag{A.7}$$

Pole at $z = z_2$ (where $z_2^2 = u - ia$, i.e. $D_2(z) = z^2 - u + ia$):

$$D_2'(z) = 2z, \qquad D_1(z_2) = z_2^2 - u - ia = (u - ia) - u - ia = -2ia. \tag{A.8}$$

Thus

$$\operatorname*{Res}_{z=z_2} f(z) = \frac{z_2^2}{(2z_2)(-2ia)} = -\frac{z_2}{4ia}. \tag{A.9}$$

The sum of the residues in the upper half-plane is therefore

$$\sum_{z_j \in \mathrm{UHP}} \operatorname*{Res}_{z=z_j} f(z) = \frac{z_1}{4ia} - \frac{z_2}{4ia} = \frac{z_1 - z_2}{4ia}. \tag{A.10}$$

Step 3: Result of the contour integral. Using Eq. (A.3), we obtain

$$I(u,a) = \int_{-\infty}^{\infty} f(t)\,dt = 2\pi i \cdot \frac{z_1 - z_2}{4ia} = \frac{\pi}{2a}\,(z_1 - z_2). \tag{A.11}$$

Now express z_1 and z_2 through u and a. With the principal branches of the square roots one has

$$z_1 = \sqrt{u + ia}, \qquad z_2 = \sqrt{u - ia}, \tag{A.12}$$

and the difference $z_1 - z_2$ can be written explicitly in terms of real square roots. Let

$$R = \sqrt{u^2 + a^2}.$$

Using the standard formula for the square root of a complex number,

$$\sqrt{u \pm ia} = \sqrt{\frac{R+u}{2}} \;\pm\; i\sqrt{\frac{R-u}{2}}, \tag{A.13}$$

we find

$$z_1 - z_2 = \left[\sqrt{\frac{R+u}{2}} + i\sqrt{\frac{R-u}{2}}\right] - \left[\sqrt{\frac{R+u}{2}} - i\sqrt{\frac{R-u}{2}}\right] = 2i\sqrt{\frac{R-u}{2}}. \tag{A.14}$$

Substituting this into Eq. (A.11) gives

$$I(u,a) = \frac{\pi}{2a} \cdot 2i\sqrt{\frac{R-u}{2}} \cdot \frac{1}{i} = \frac{\pi}{a}\sqrt{\frac{R-u}{2}}, \qquad R = \sqrt{u^2 + a^2}. \tag{A.15}$$

(Here the factor $1/i$ comes from the fact that $a > 0$ and the chosen contour corresponds to the retarded prescription; the final result is real and positive, as required for an integral of a positive function.)

Thus we have obtained a fully explicit, closed-form expression for the integral

$$I(u,a) = \int_{-\infty}^{+\infty} \frac{t^2}{(t^2 - u)^2 + a^2}\,dt = \frac{\pi}{a}\sqrt{\frac{\sqrt{u^2 + a^2} - u}{2}}. \tag{A.16}$$

When this result is substituted back into the prefactors arising from the k-space integration in the effective-mass model, one recovers exactly the broadened DOS formula of the form

$$\rho(E) \propto \sqrt{\frac{\sqrt{(E - E_c^*)^2 + (\Gamma/2)^2} + (E - E_c^*)}{2}}, \tag{A.17}$$

after a straightforward rearrangement of u and a in terms of $E - E_c^*$ and Γ (and accounting for the additional factors of $(2m^*/\hbar^2)^{3/2}$).

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